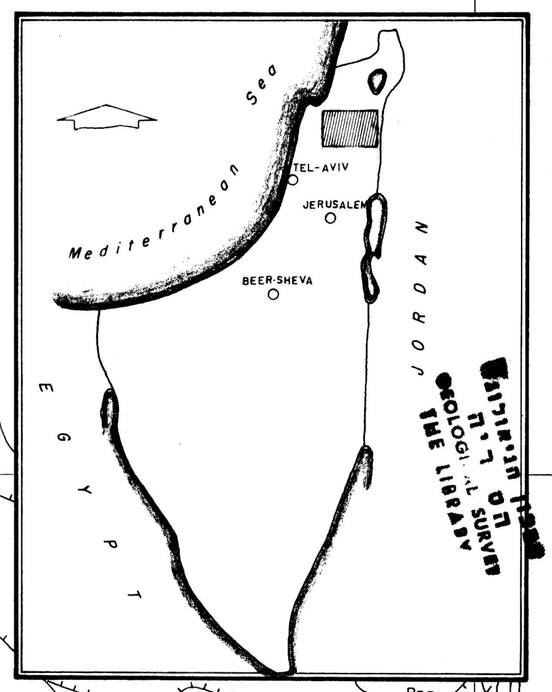
ST(569.4) RA

GEOTHERMAL PROJECT

ELECTRICAL RESISTIVITY SURVEY

THE YIZREEL AND HAROD VALLEYS

March 1977 GT/149/74 (2)



GEOLOGICAL SURVEY
THE LIBRARY

ALLONIM
YOQNE'AM
MISHMAR-HAEMEQ B
SARID 1
GIDO'N 1
GIDO'N 3
GIDO'N 4
GIDO'N 5
AFULA
DEVORAH 1
463-1
463-10
416-84
416-85
416-86
GID'ONA
EN-SHOQEQ
738-12
738-11
738-9
PROFILE 1
PROFILE 2
PROFILE 3
PROFILE 4
PROFILE 5

QUATERNARY - RECENT	q	EOCENE	e	FAULT	
UPPER PLEISTOCENE - RECENT	qt	SENONIAN - PALEOCENE UNDIVIDED	S	SOURCE	(32)
UPPER PLEISTOCENE (Lisan fm.)	ql	TURONIAN	t	MODIFIED SHLUMBERGER SOUNDING	→
LOWER PLEISTOCENE (Melanopsis beds)	qm	UPPER CENOMANIAN - TURONIAN	C3t	MONOPOLE SOUNDING	--→
NEOGENE - QUATERNARY (Alluvial fans)	qn	UPPER CENOMANIAN	C2	DIPOLE-DIPOLE PROFILE	3 4 5
PLIOCENE MARINE	p	LOWER CENOMANIAN	C1	SHLUMBERGER SOUNDING	463-10
MIOCENE MARINE	m	LOWER CRETACEOUS MARINE	Lc	WELL	
NEOGENE MOSTLY CONTINENTAL	nc	NEOGENE - QUATERNARY (Volcanics)	βnq	SPRING	
OLIGOCENE	ol.	UPPER CRETACEOUS (Volcanics)	βc		

THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS - ISRAEL

LOCATION and GEOLOGICAL MAP

EMEQ YIZREEL

Scale

km. 0 0.5 1 2 3 4 5 6 km.

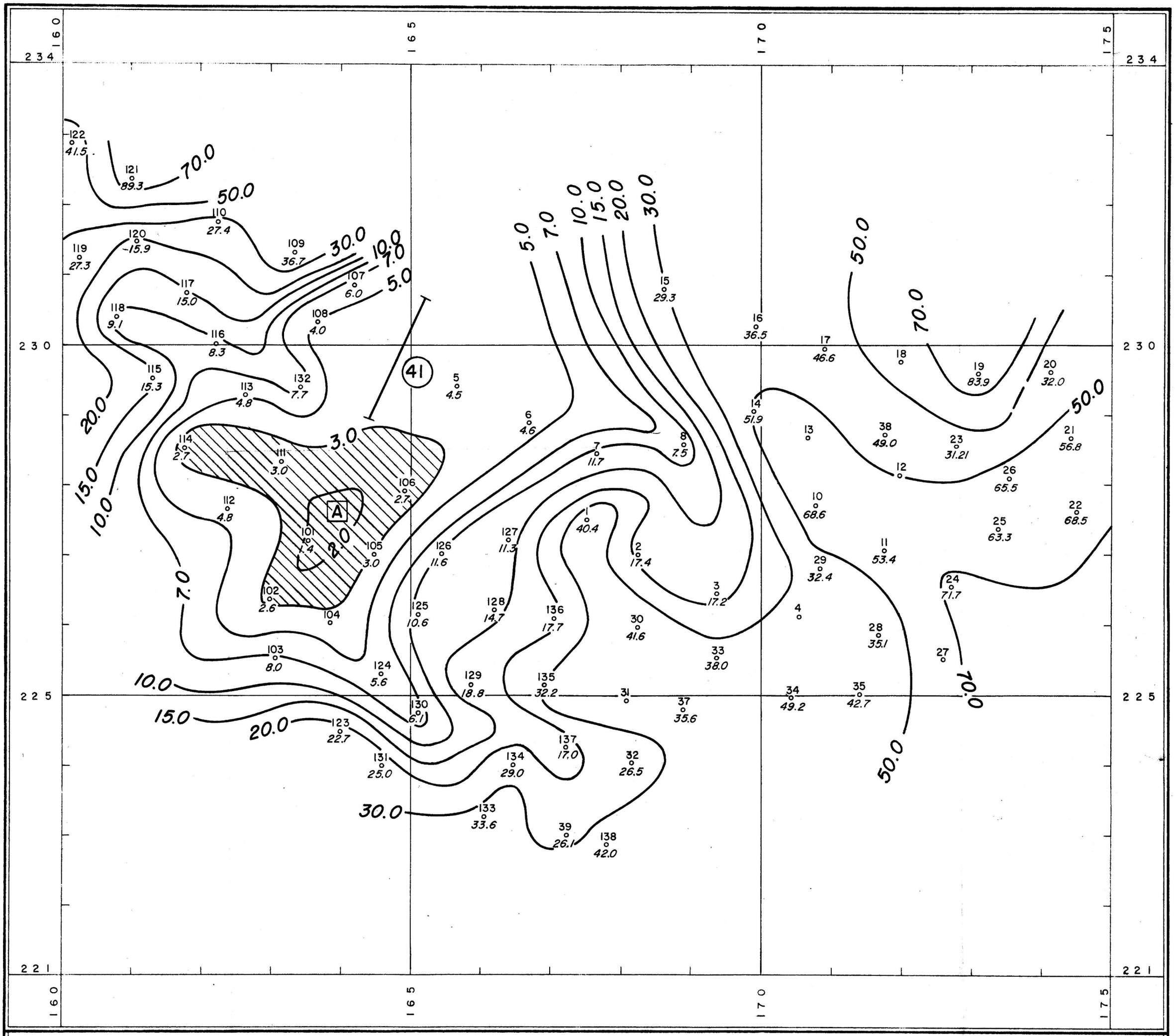
GEOLOGY:

From GEOLOGICAL MAP of ISRAEL, Scale 1:250000

Date: March 1977

GE/149/74 (53964)

Figure 1



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ROVING DIPOLE
APPARENT RESISTIVITY MAP
SOURCE 41
EMEQ YIZREEL

SCALE

km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

January 1977

GT/149/74 (52721)

Figure 3

THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS – ISRAEL

ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 42

EMEQ YIZREEL

SCALE

km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

January 1977

GT/149/74 (52722)

Figure 4

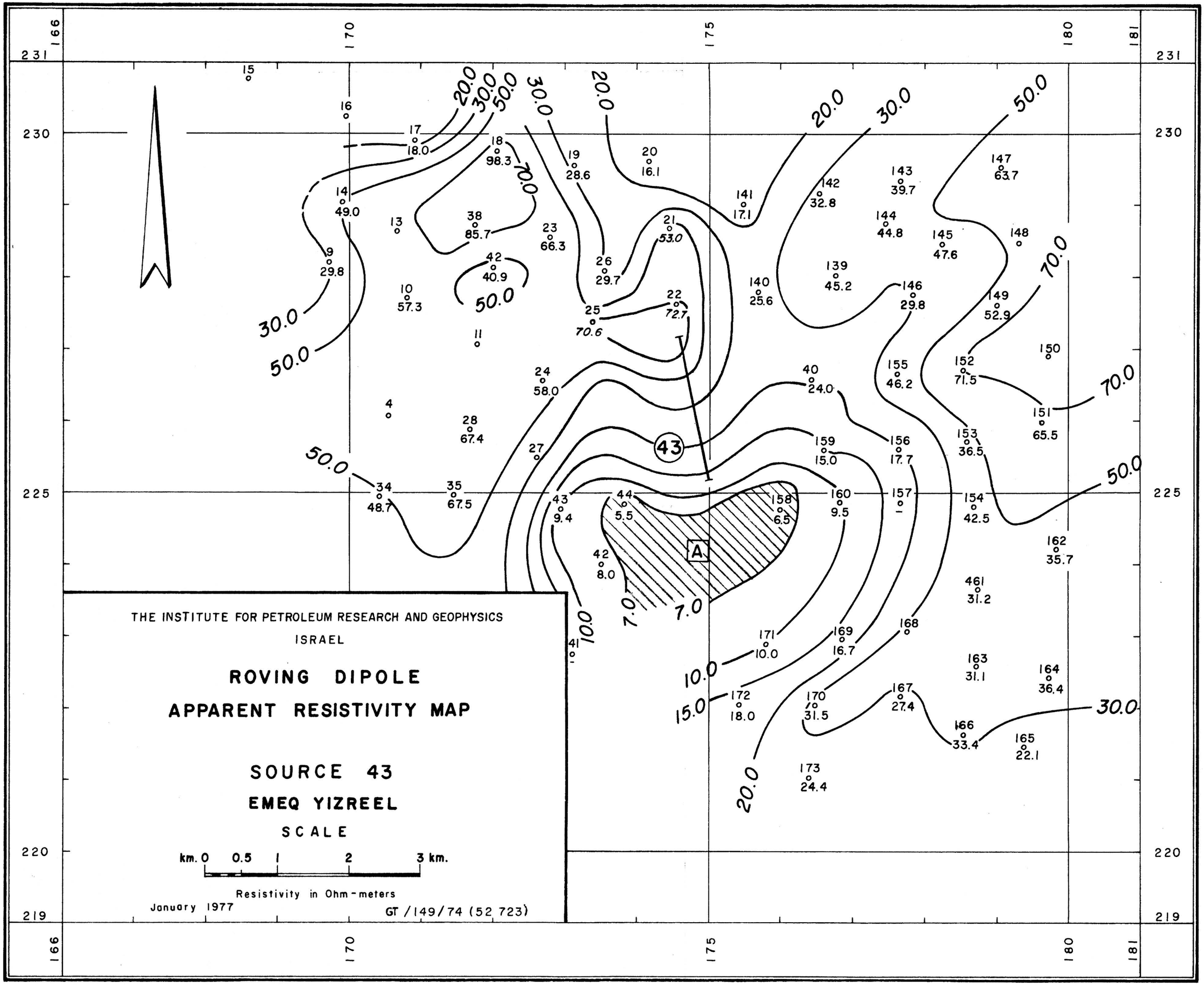
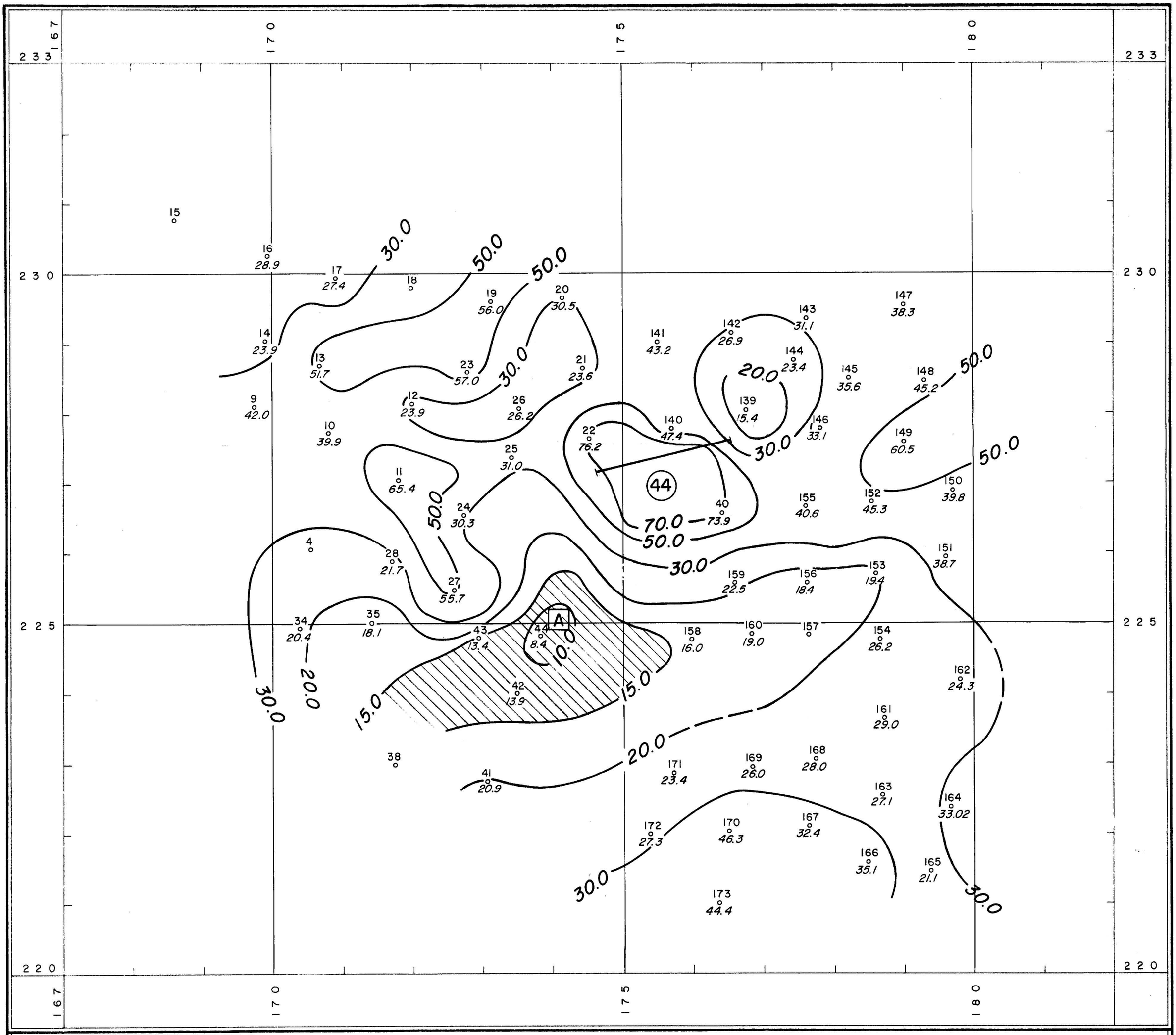

166
170
175
180
181
231
230
225
220
219
THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS
ISRAEL
ROVING DIPOLE
APPARENT RESISTIVITY MAP
SOURCE 43
EMEQ YIZREEL
SCALE
km. 0 0.5 1 2 3 km.
Resistivity in Ohm-meters
January 1977
GT/149/74 (52 723)
43
A

Figure 5



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 44

EMEQ YIZREEL

SCALE

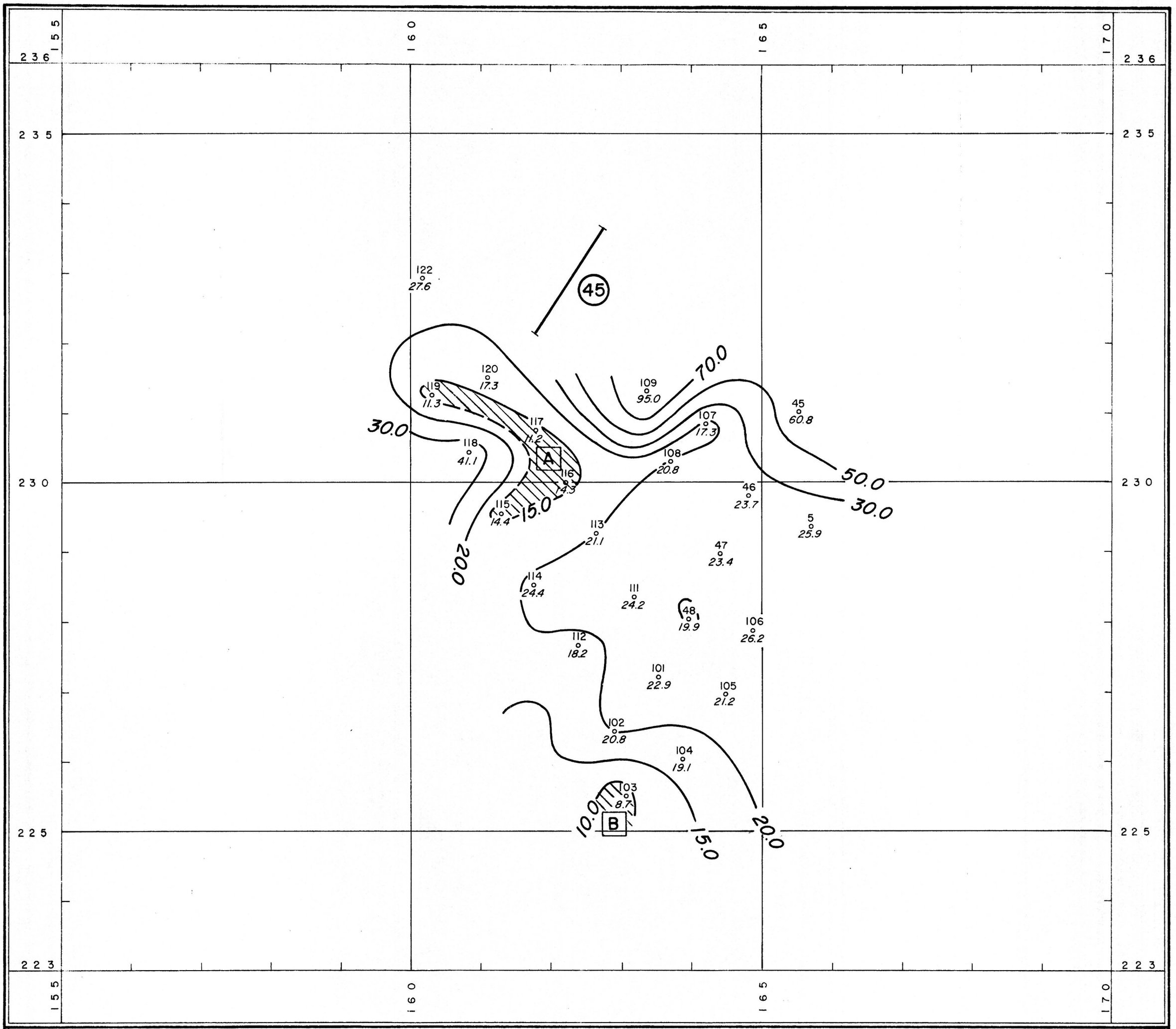
km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

January 1977

GT/149/74 (52886)

Figure 6



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 45

EMEQ YIZREEL

SCALE

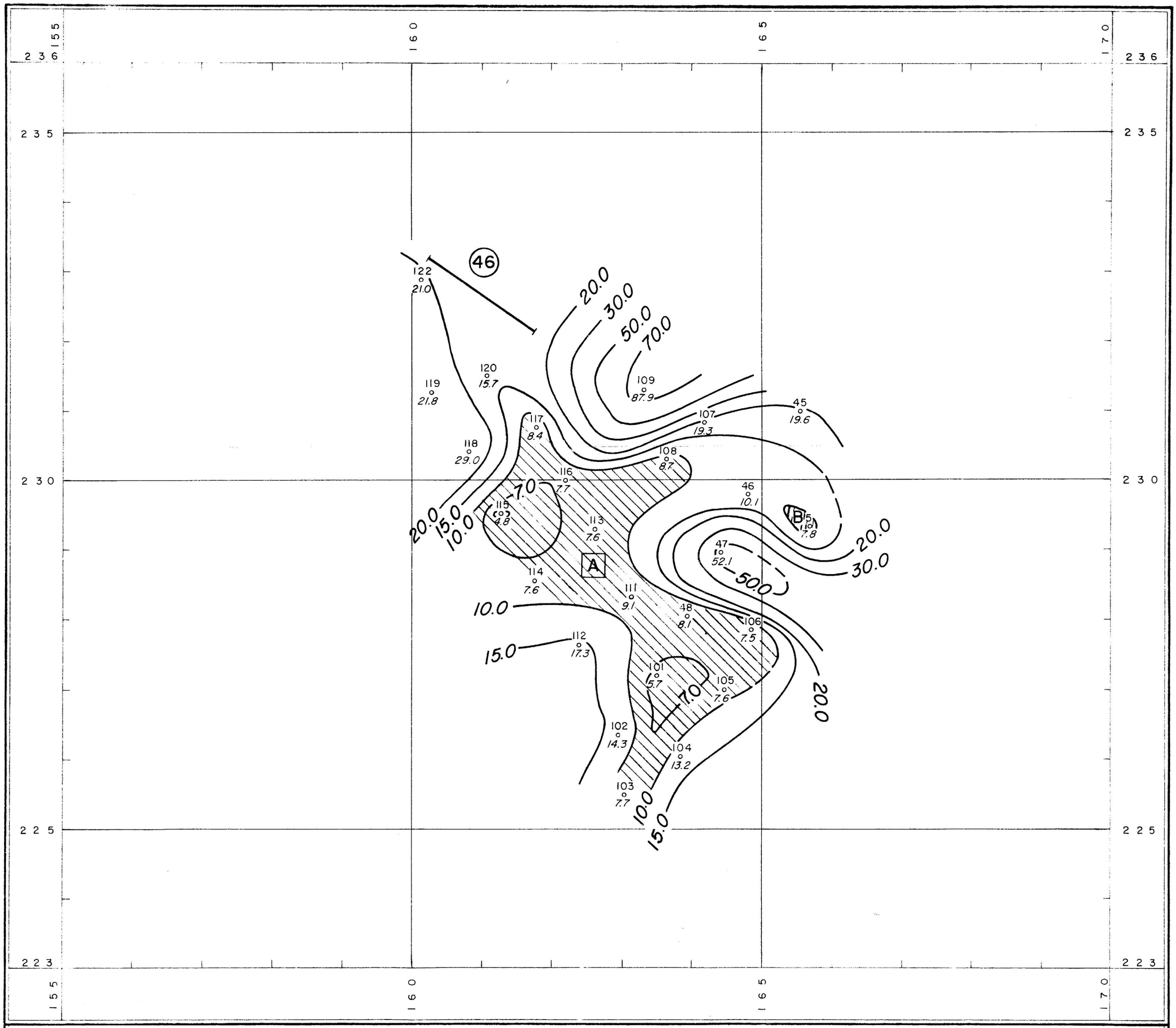
km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

January 1977

GT/149/74 (52887)

Figure 7



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 46

EMEQ YIZREEL

SCALE

km. 0 0.5 1 2 3 km.

January 1977

Resistivity in Ohm-meters

GT/149/74 (52 888)

Figure 8

THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS – ISRAEL

ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 47

EMEQ YIZREEL

SCALE

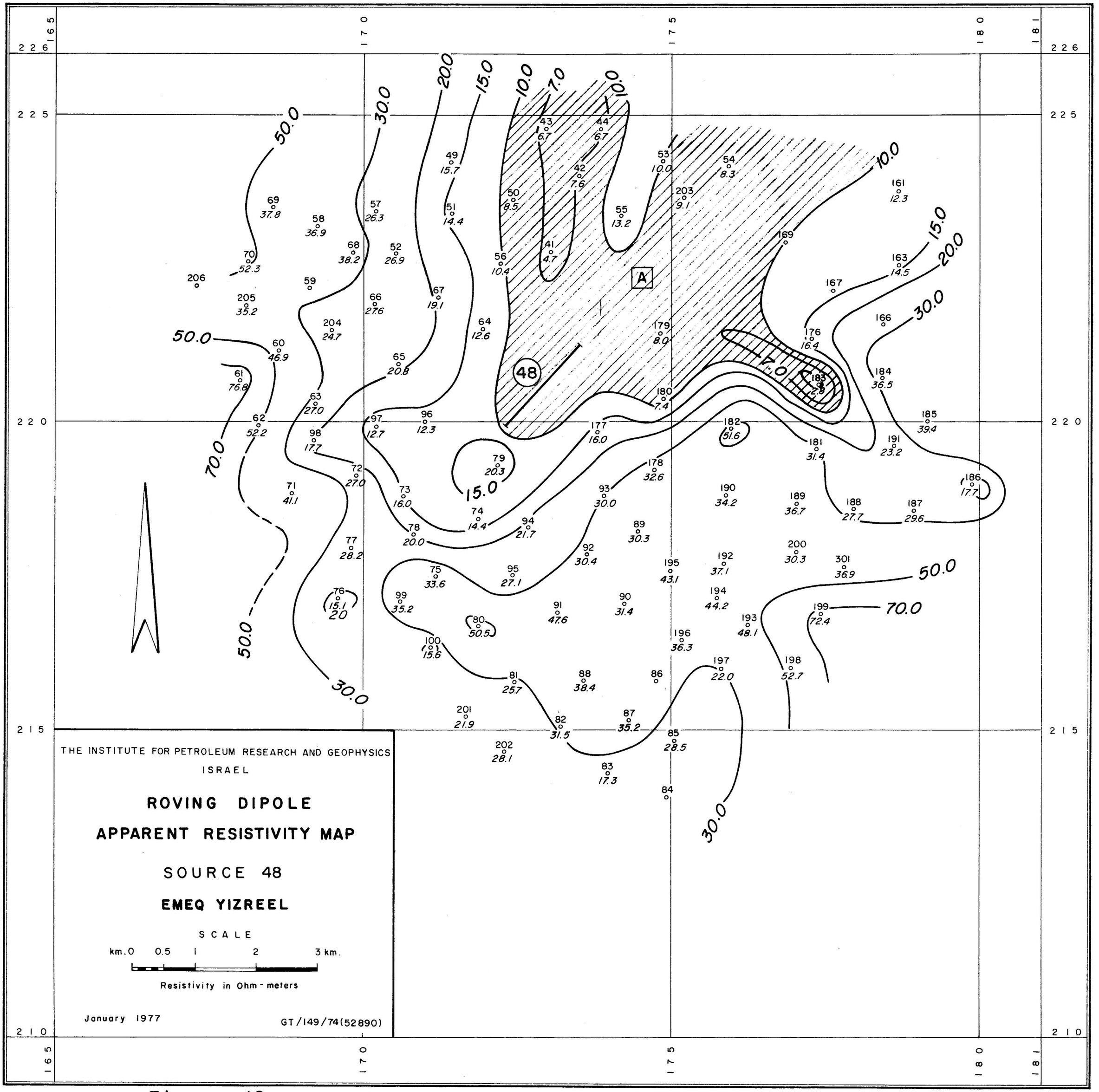
km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

January 1977

GT/149/74 (52889)

Figure 9


165
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48
A
50.0
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20.0
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7.0
70.0
THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS
ISRAEL
ROVING DIPOLE
APPARENT RESISTIVITY MAP
SOURCE 48
EMEQ YIZREEL
SCALE
km. 0 0.5 1 2 3 km.
Resistivity in Ohm-meters
January 1977
GT/149/74(52890)

Figure 10

THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

ISRAEL

ROVING DIPOLE
APPARENT RESISTIVITY MAP

SOURCE 49

EMEQ YIZREEL

SCALE

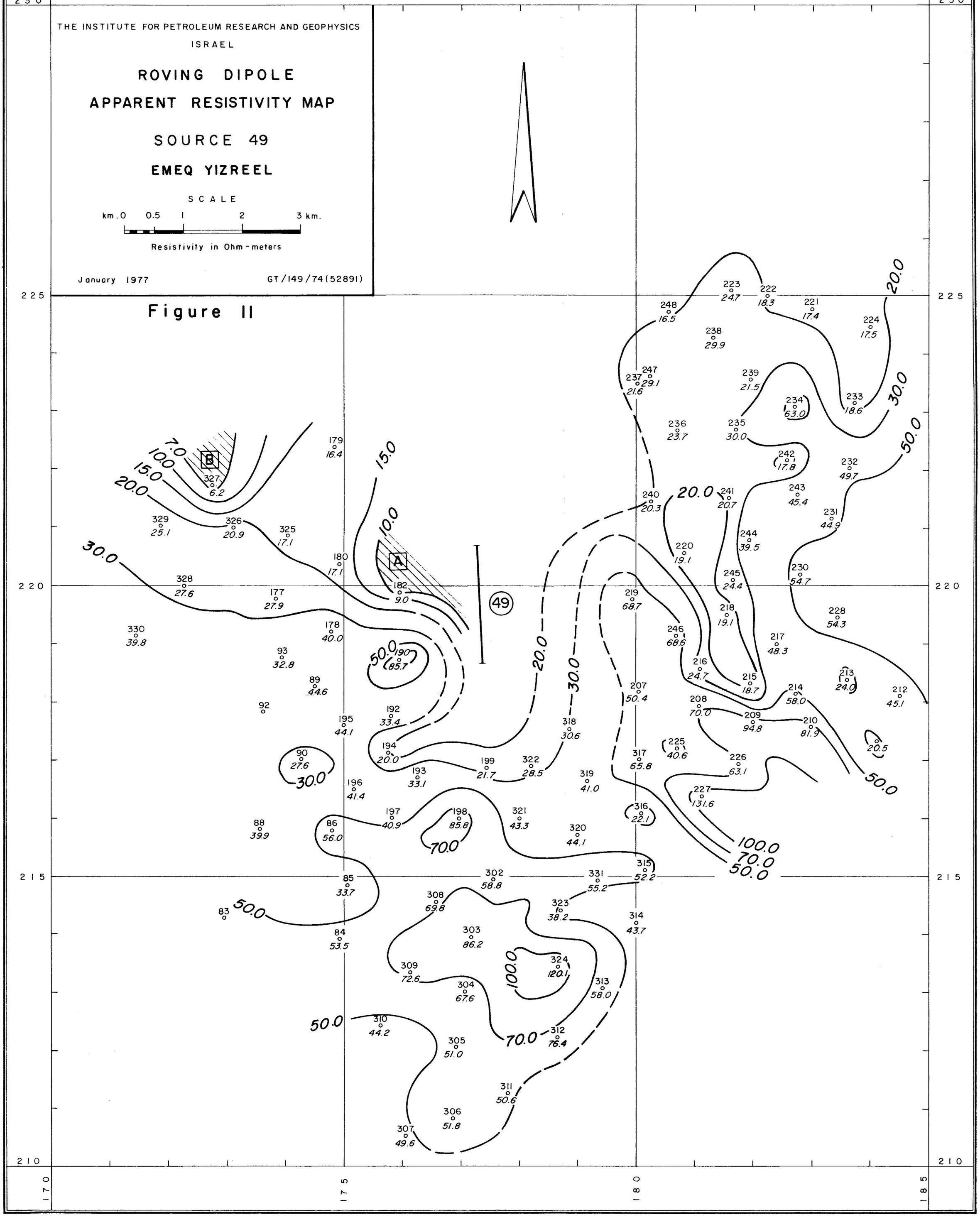
km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

January 1977

GT/149/74(52891)

Figure 11



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 50

EMEQ YIZREEL

SCALE

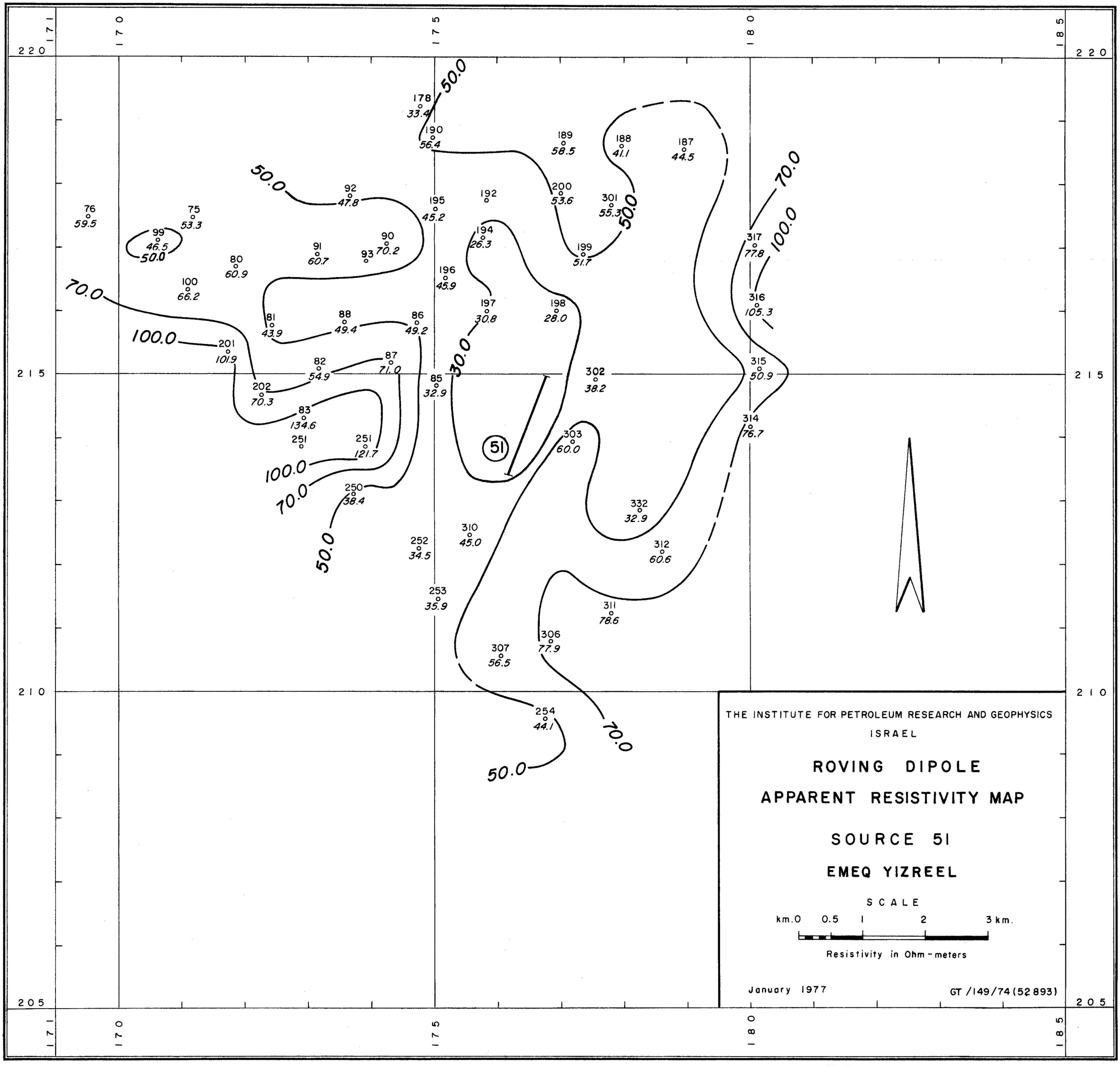
km. 0 0.5 1 2 3 km.

January 1977

Resistivity in Ohm-meters

GT/149/74 (52892)

Figure 12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301 55.3
199 51.7
76 59.5
75 53.3
99 46.5
50.0
91 60.7
90 70.2
93
80 60.9
100 66.2
196 45.9
194 26.3
197 30.8
198 28.0
317 77.8
316 105.3
81 43.9
88 49.4
86 49.2
201 101.9
82 54.9
87 71.0
202 70.3
85 32.9
302 38.2
315 50.9
314 76.7
83 134.6
251
251 121.7
303 60.0
51
250 38.4
332 32.9
310 45.0
252 34.5
312 60.6
253 35.9
311 78.6
306 77.9
307 56.5
254 44.1
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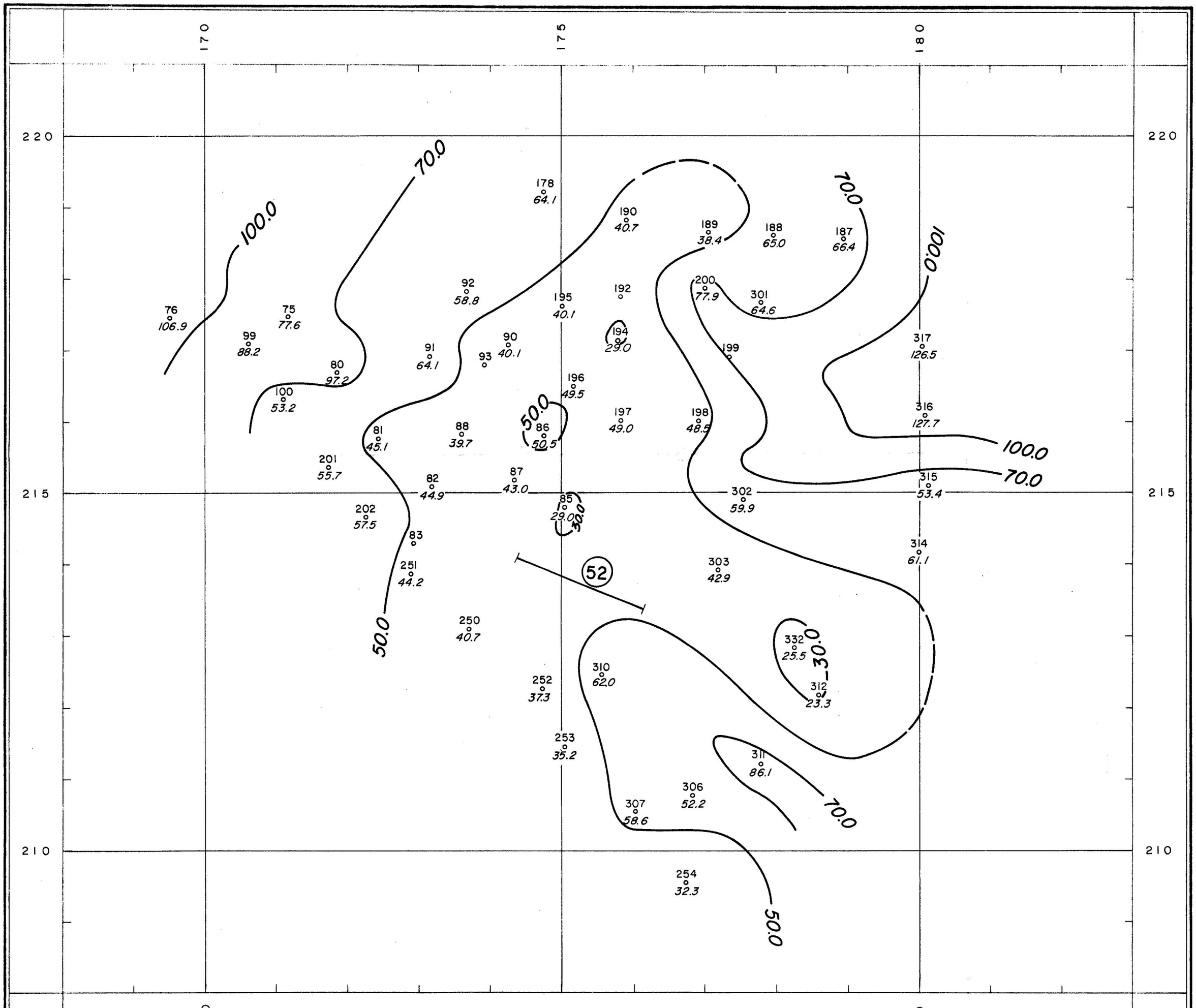
ISRAEL
ROVING DIPOLE
APPARENT RESISTIVITY MAP
SOURCE 51
EMEQ YIZREEL
SCALE
km.0 0.5 1 2 3 km.
Resistivity in Ohm-meters
January 1977
GT/149/74(52893)

Figure 13



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 52

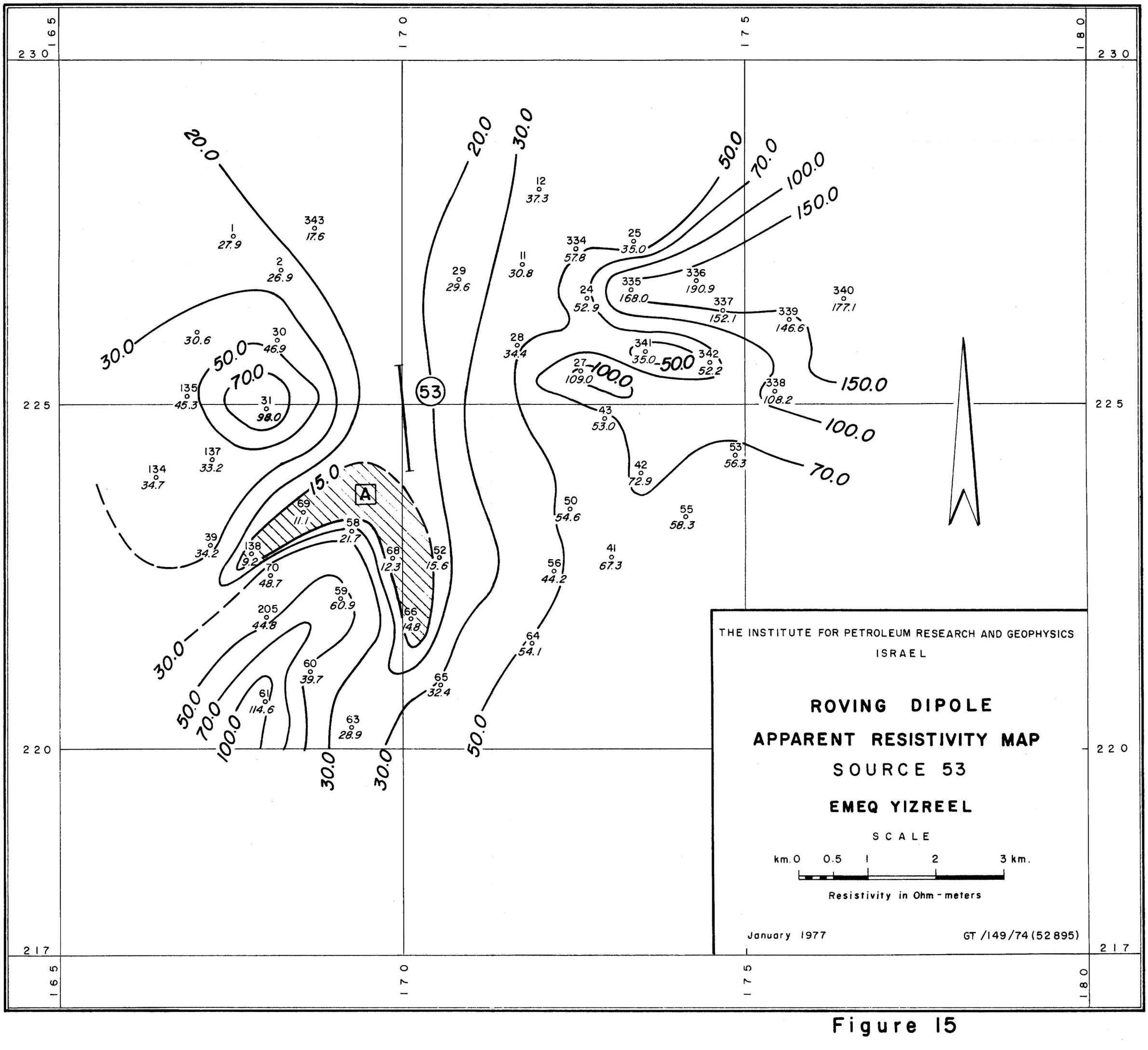
EMEQ YIZREEL

SCALE

km. 0 0.5 1 2 3 km.

January 1977 Resistivity in Ohm-meters GT/149/74 (52894)

Figure 14


165
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100.0
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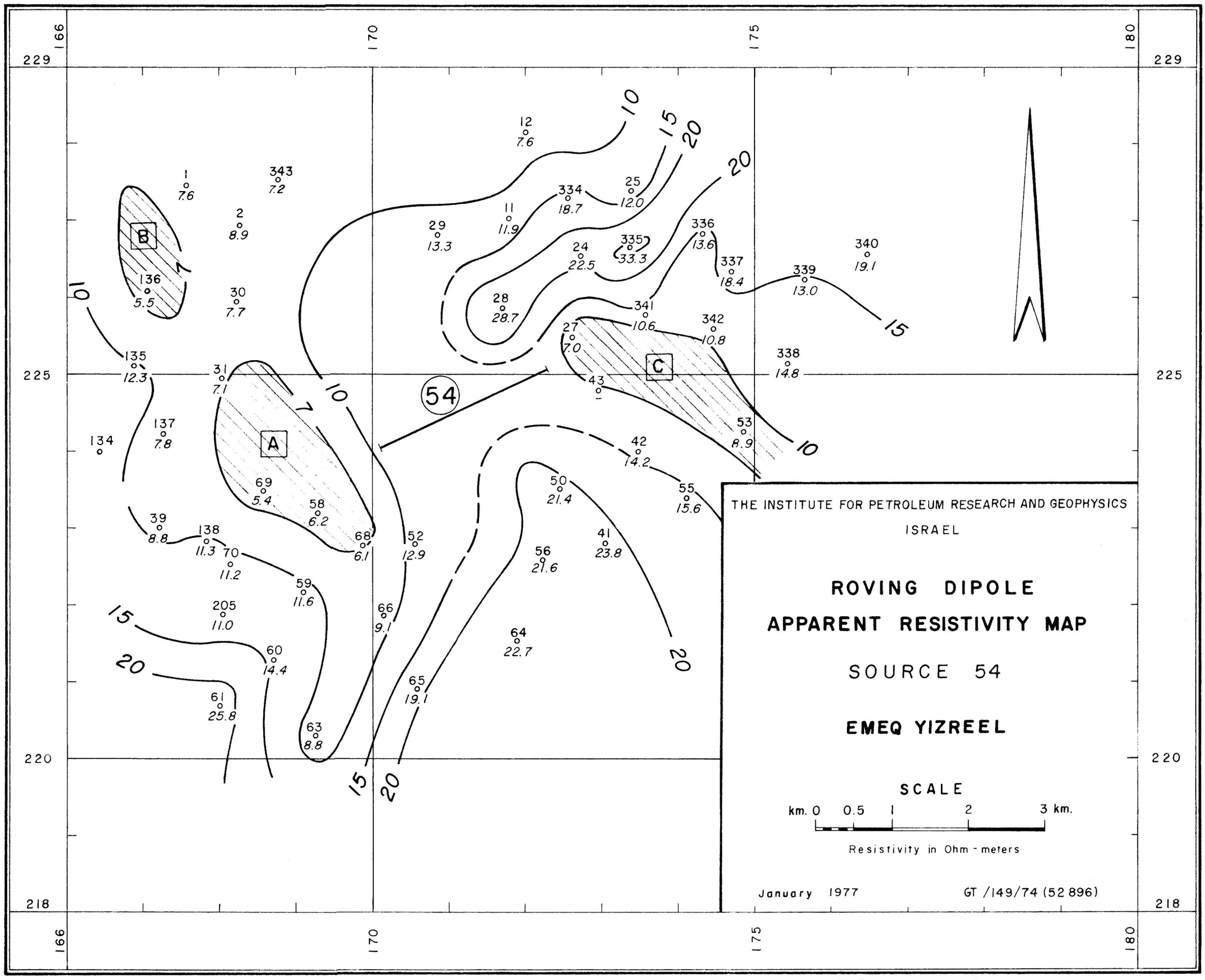
THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS
ISRAEL
ROVING DIPOLE
APPARENT RESISTIVITY MAP
SOURCE 53
EMEQ YIZREEL
SCALE
km. 0 0.5 1 2 3 km.
Resistivity in Ohm-meters
January 1977
GT/149/74(52895)

Figure 15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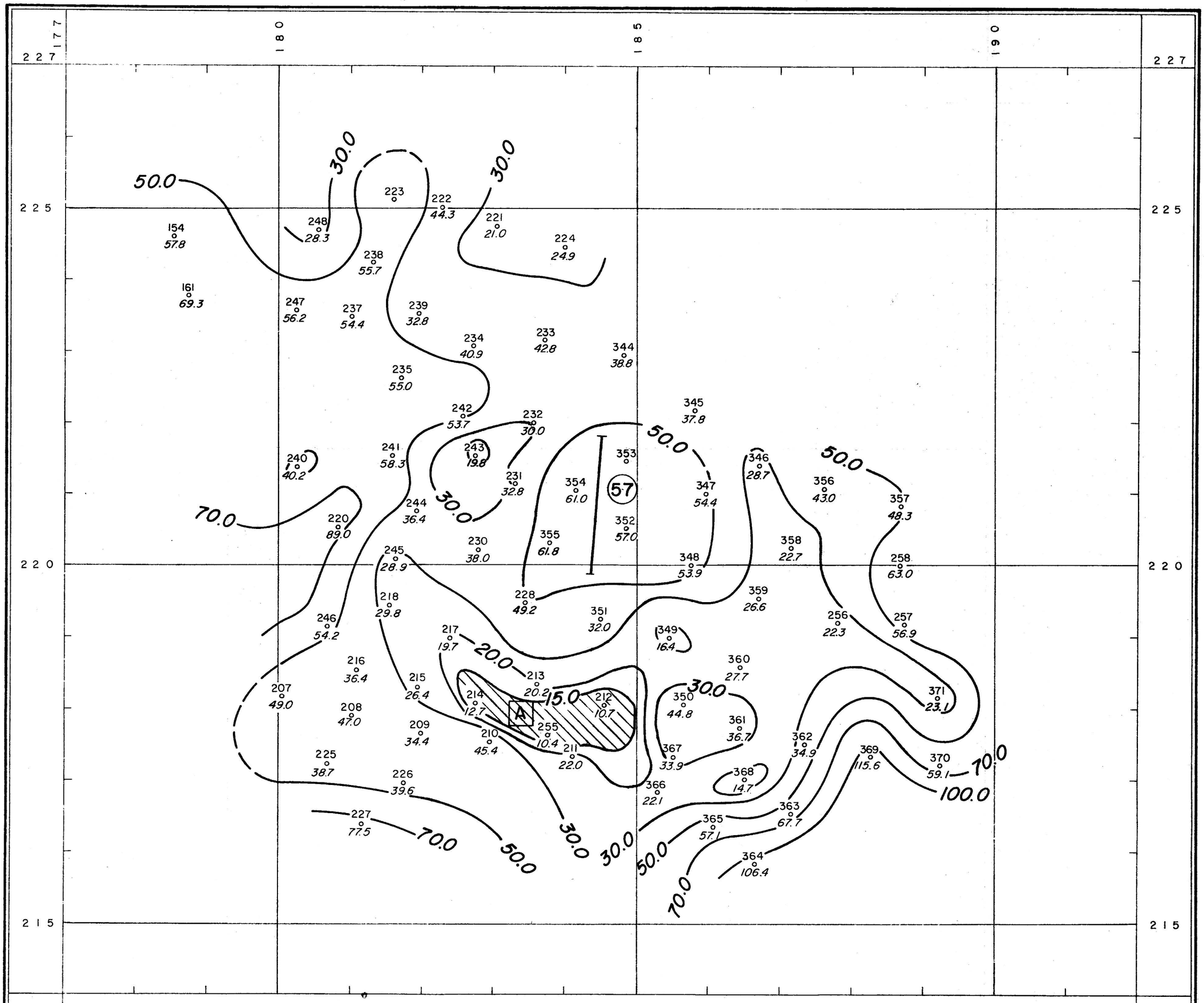
THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS
ISRAEL
ROVING DIPOLE
APPARENT RESISTIVITY MAP
SOURCE 54
EMEQ YIZREEL
SCALE
km. 0 0.5 1 2 3 km.
Resistivity in Ohm - meters
January 1977
GT /149/74 (52 896)

Figure 16



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ROVING DIPOLE
APPARENT RESISTIVITY MAP
SOURCE 57
EMEQ YIZREEL

SCALE

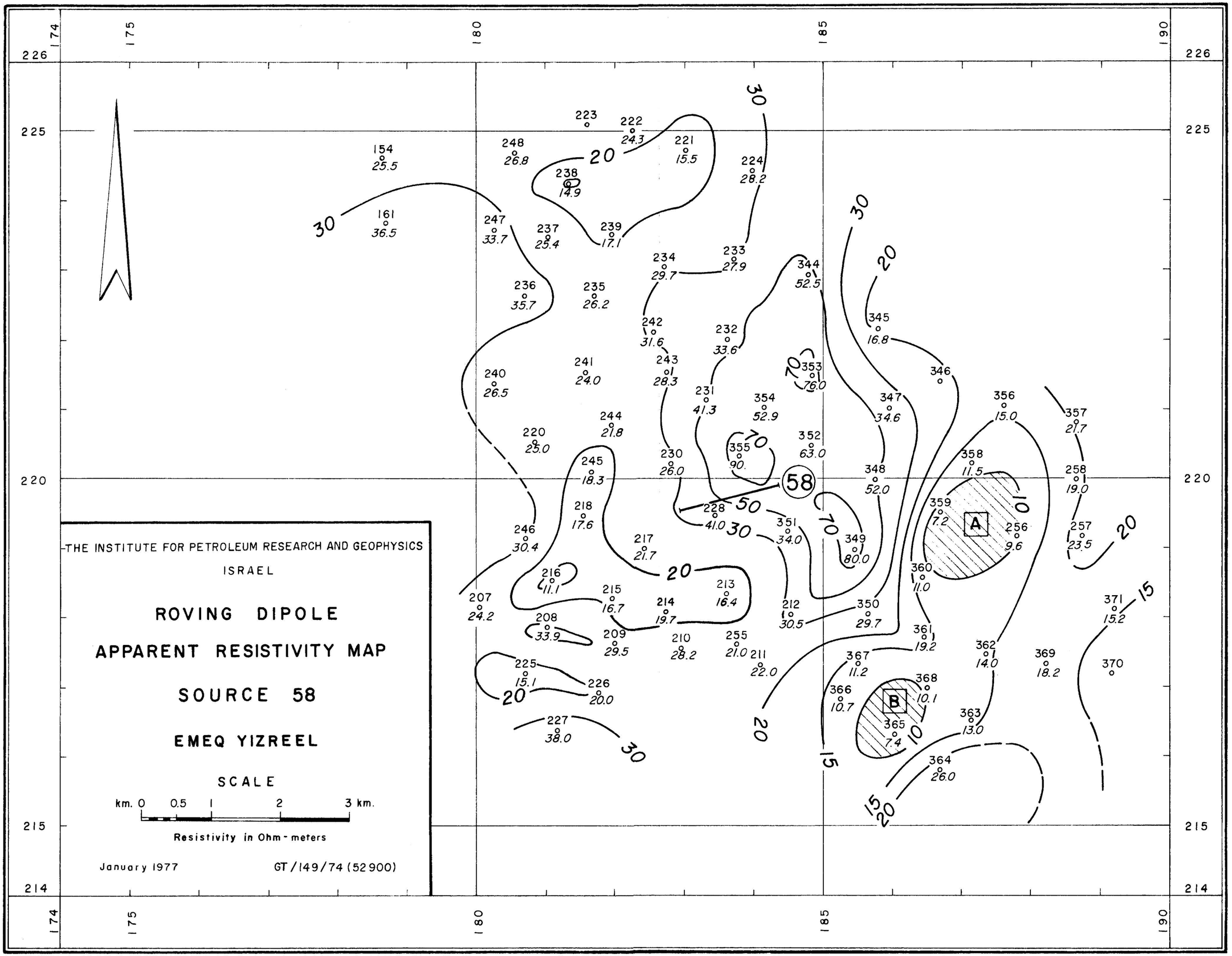
km. 0 0.5 1 2 3 km.

January 1977

Resistivity in Ohm-meters

GT/149/74 (52899)

Figure 17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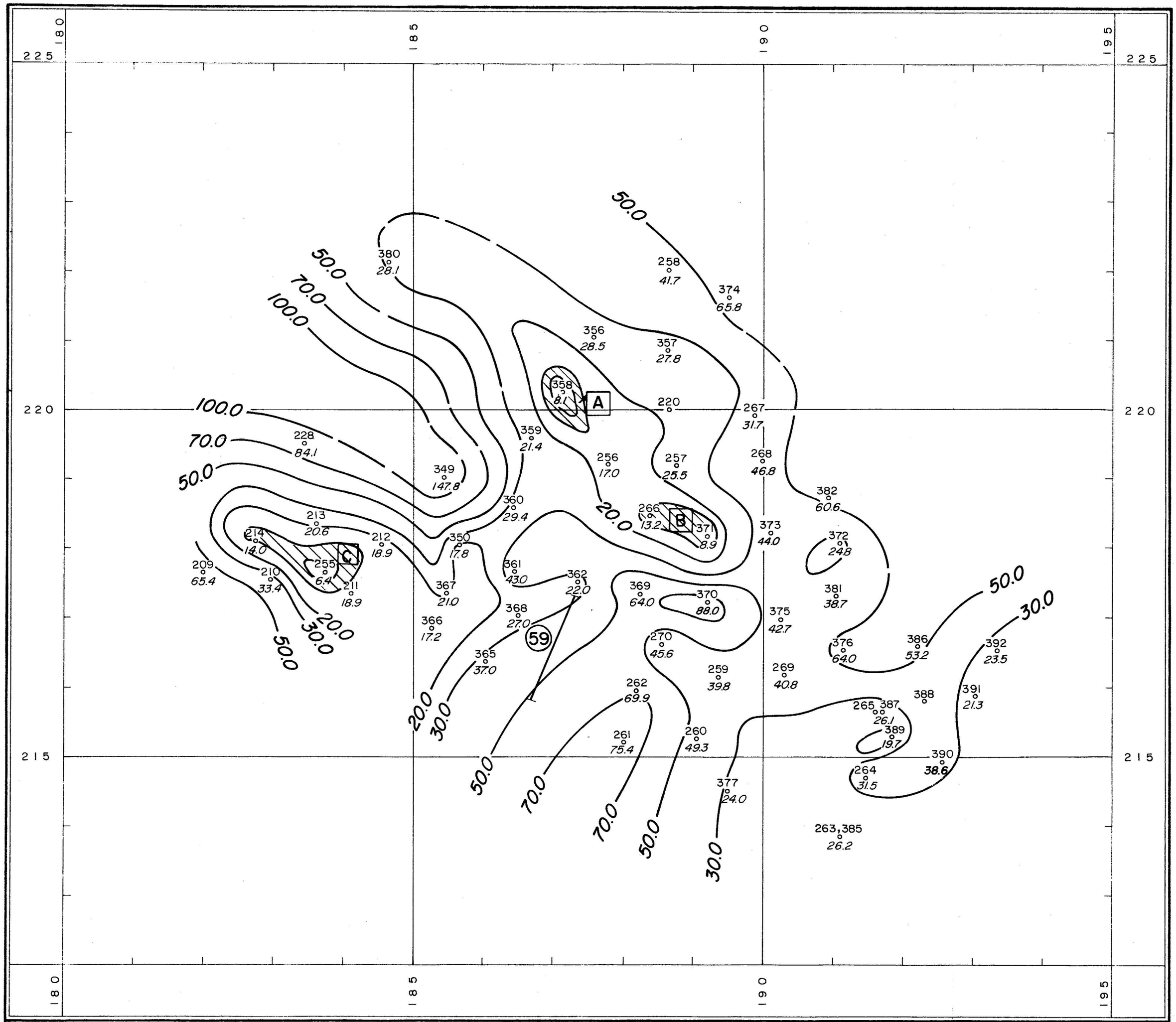
15
58
THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS
ISRAEL
ROVING DIPOLE
APPARENT RESISTIVITY MAP
SOURCE 58
EMEQ YIZREEL
SCALE
km. 0 0.5 1 2 3 km.
Resistivity in Ohm-meters
January 1977
GT/149/74 (52900)

Figure 18



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 59

EMEQ YIZREEL

SCALE

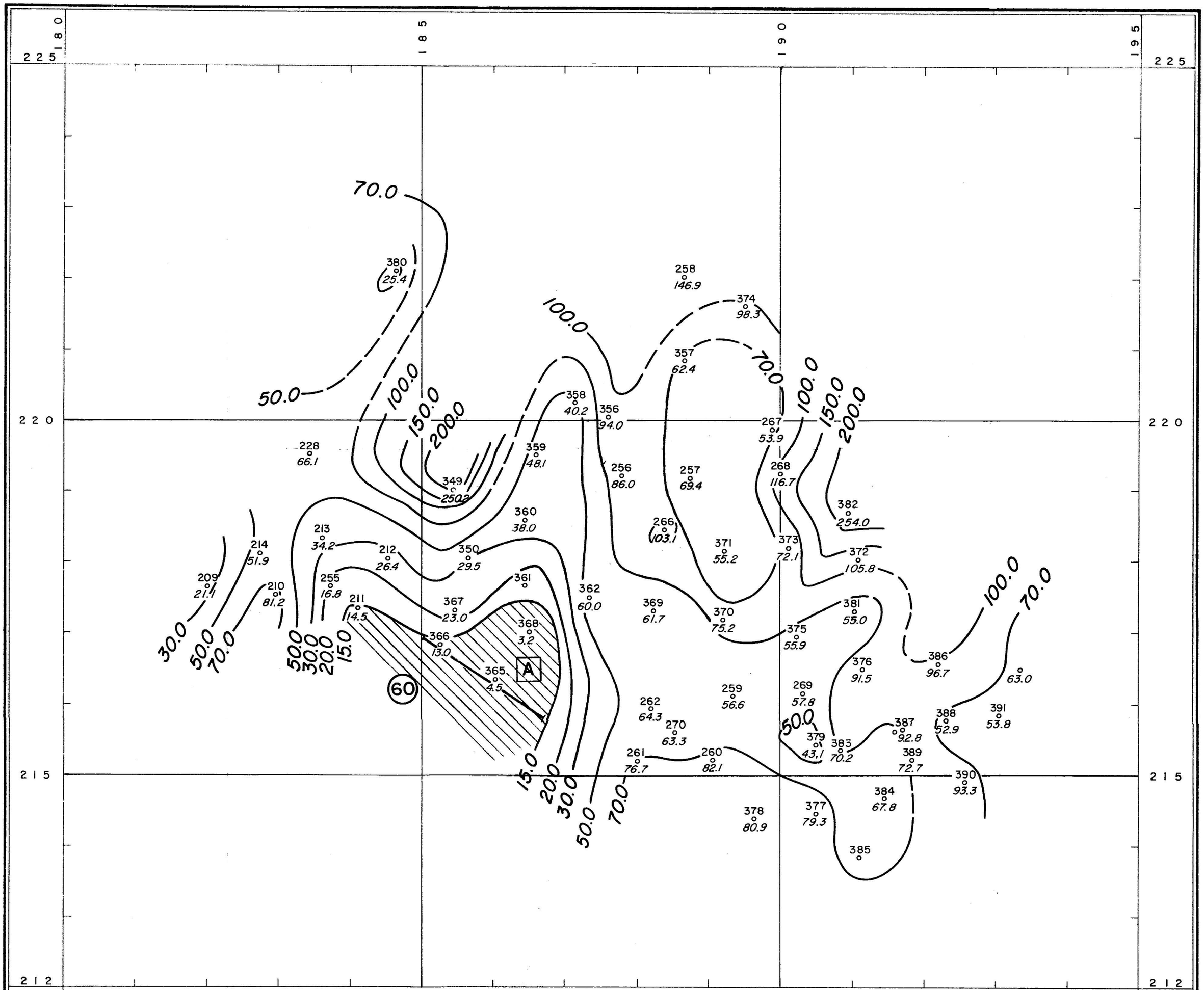
km. 0 0.5 1 2 3 km.

January 1977

Resistivity in Ohm-meters

GT/149/74 (52901)

Figure 19



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 60

EMEQ YIZREEL

SCALE

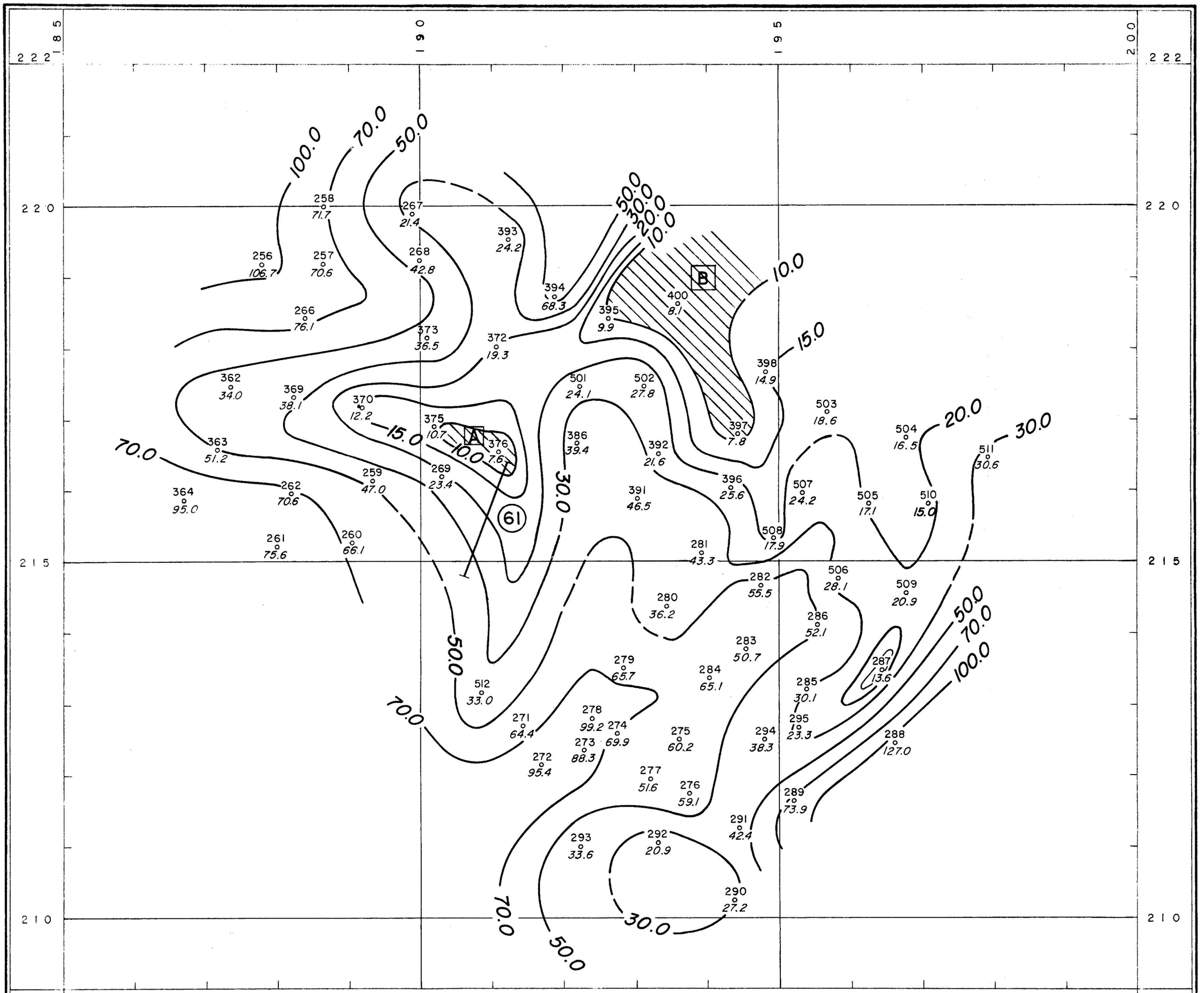
km. 0 0.5 1 2 3 km.

January 1977

Resistivity in Ohm-meters

GT/149/74 (52902)

Figure 20



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 61

EMEQ YIZREEL

SCALE

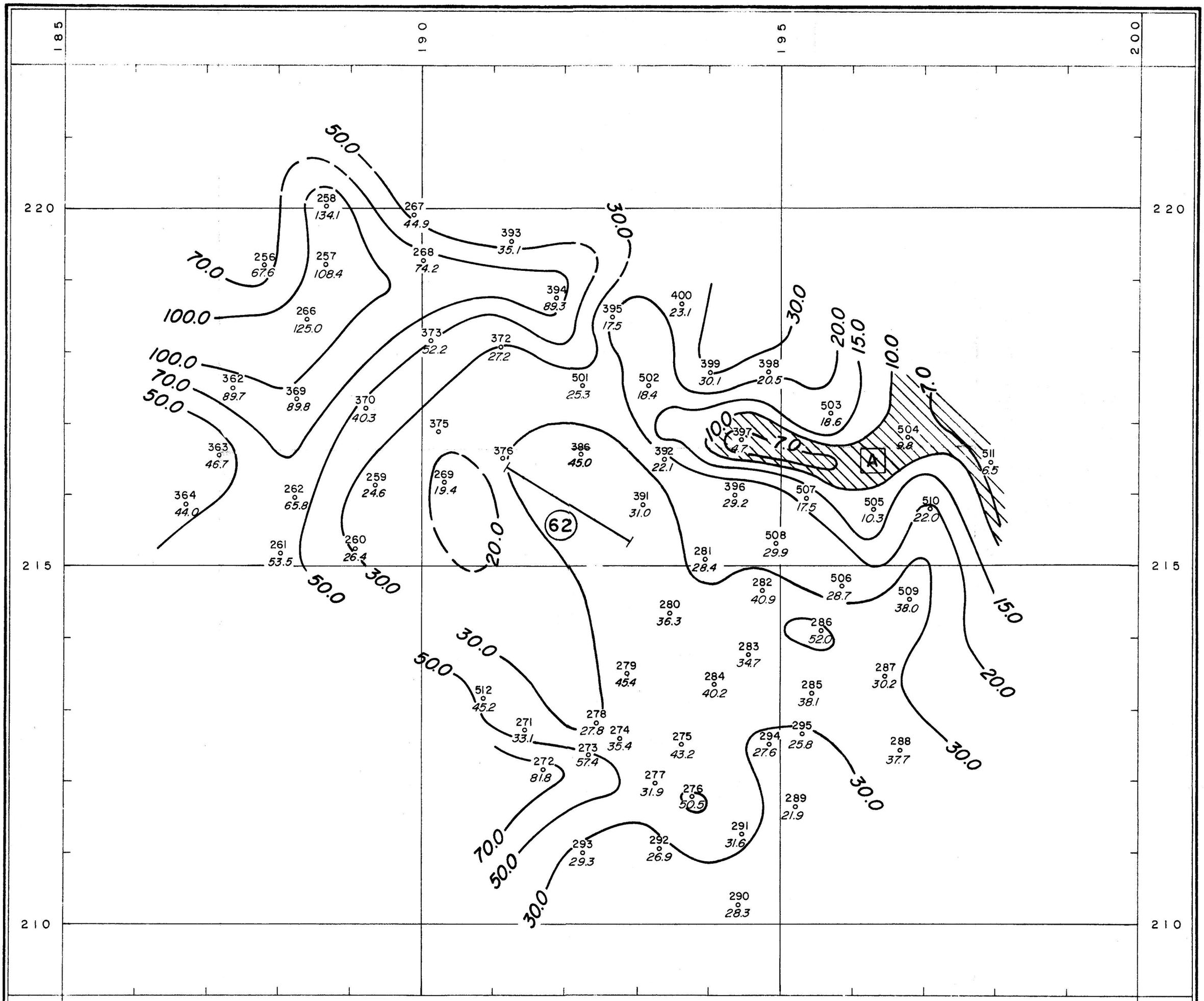
km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

January 1977

GT/149/74 (52926)

Figure 21



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 62

EMEQ YIZREEL

SCALE

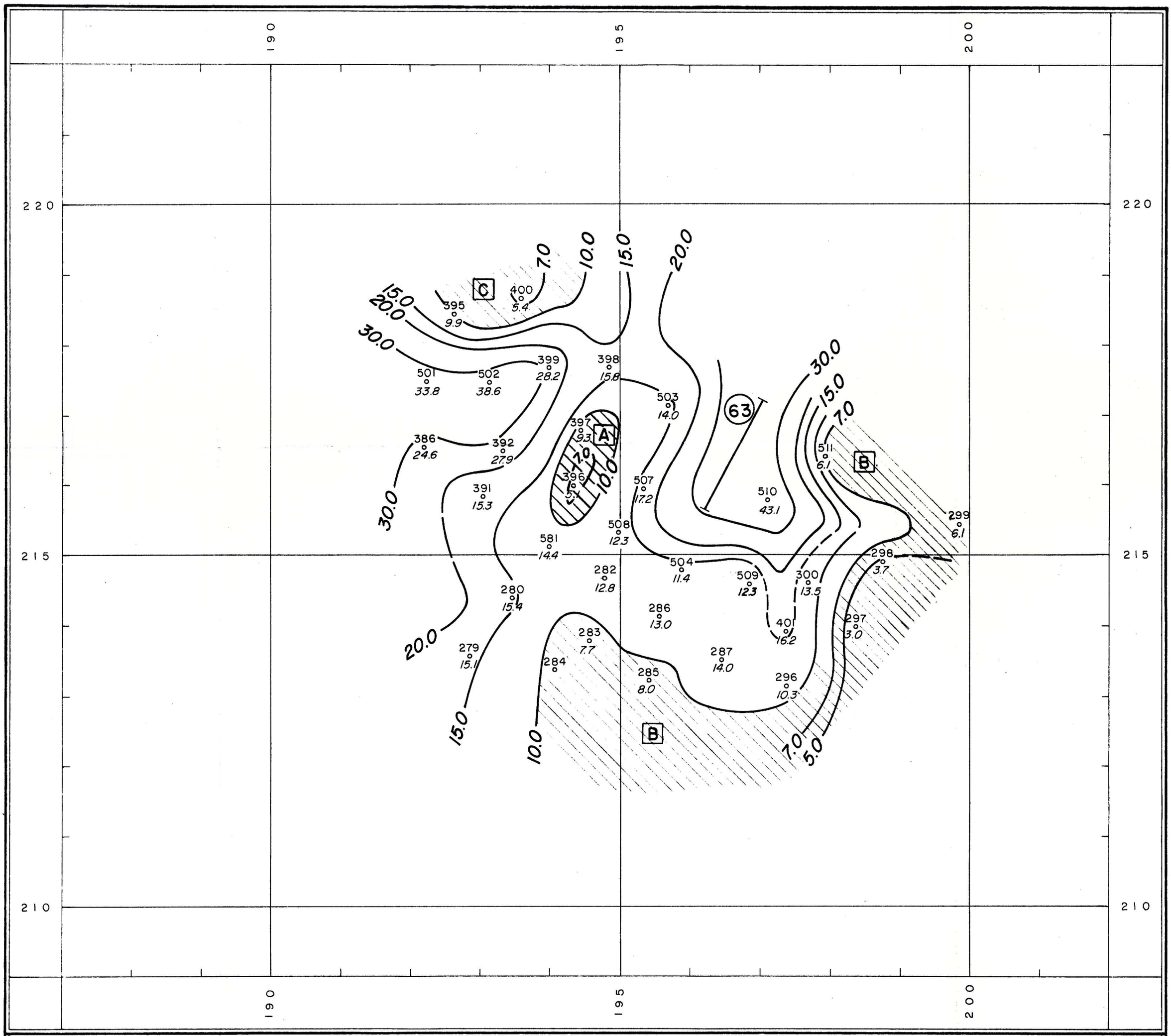
km. 0 0.5 1 2 3 km.

January 1977

Resistivity in Ohm-meters

GT/149/74 (52927)

Figure 22



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 63

EMEQ YIZREEL

SCALE

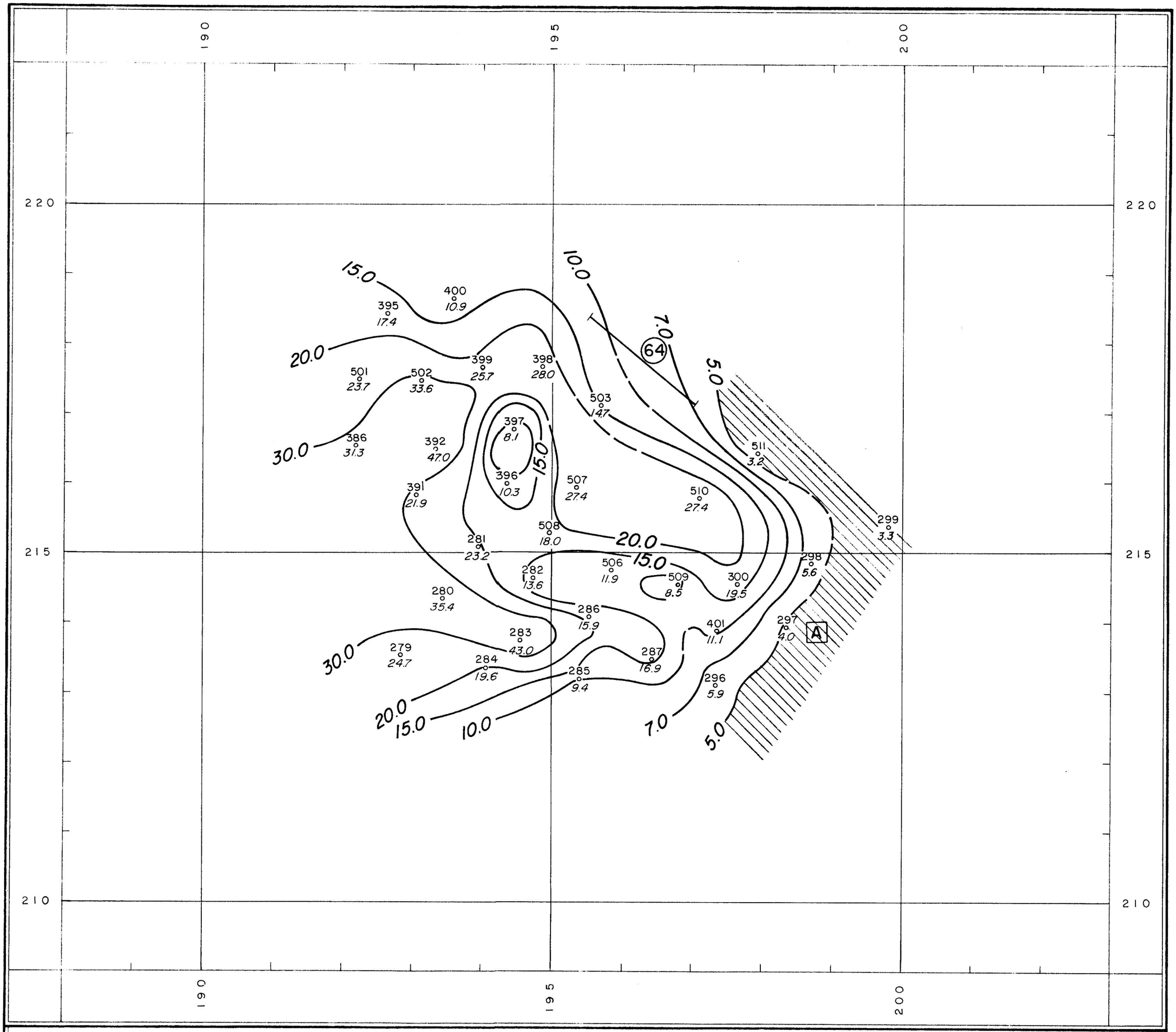
km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

January 1977

GT/149/74 (52928)

Figure 23



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 64

EMEQ YIZREEL

SCALE

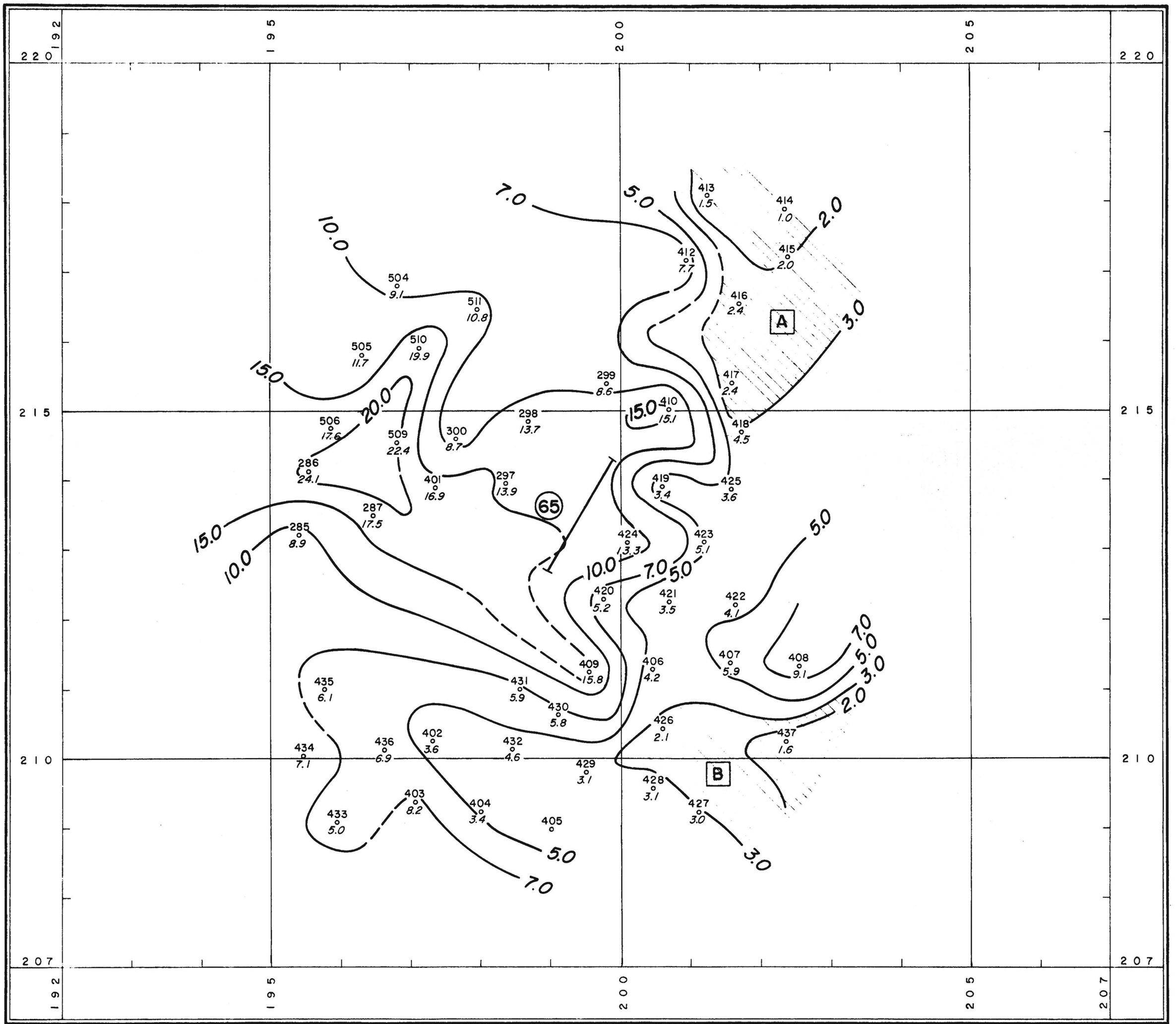
km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

January 1977

GT/149/74 (52929)

Figure 24



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 65

EMEQ YIZREEL - BET SHE'AN

SCALE

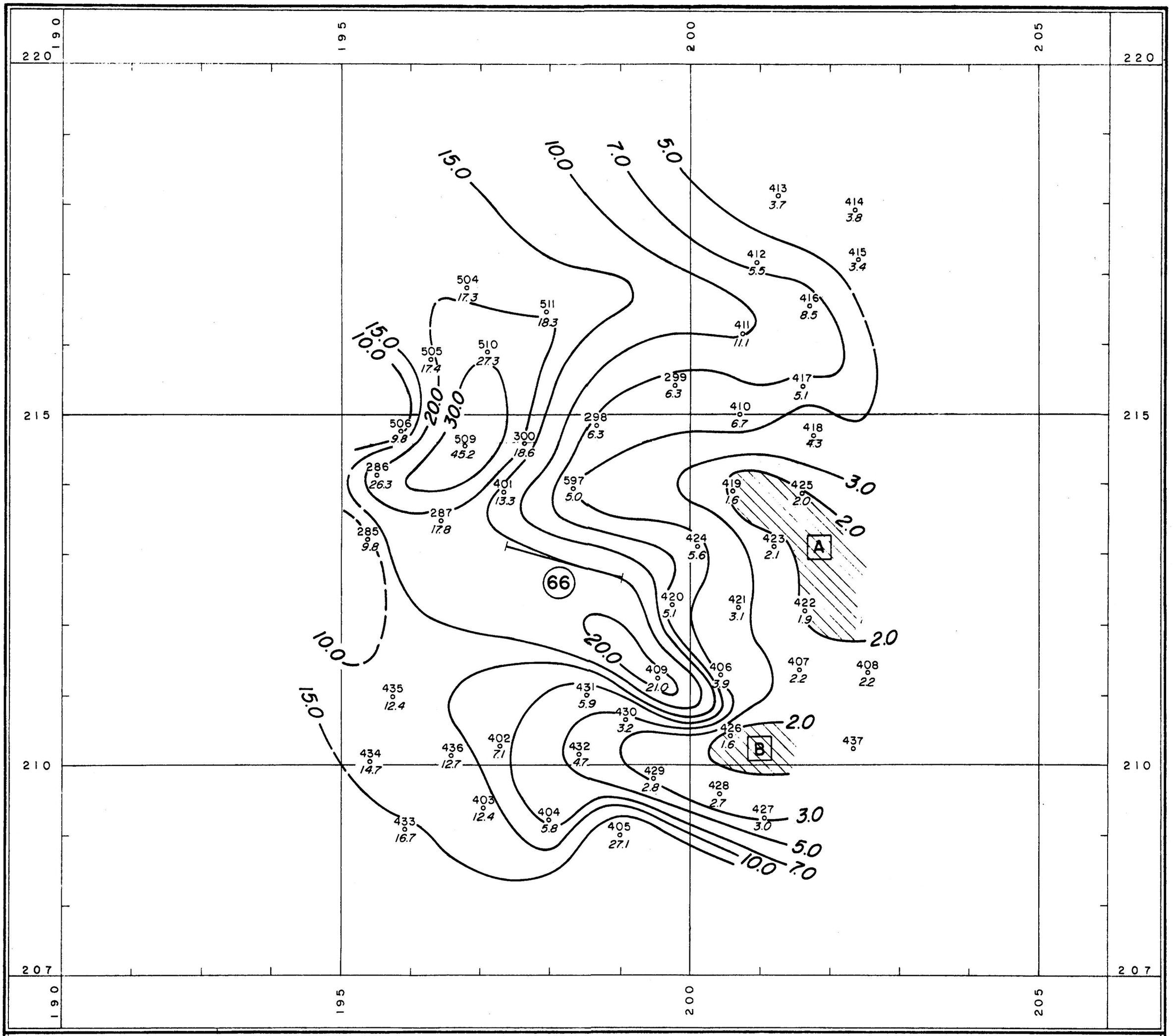
km. 0 0.5 1 2 3 km.

January 1977

Resistivity in Ohm-meters

GT/149/74 (52930)

Figure 25



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 66

EMEQ YIZREEL - BET SHE'AN

SCALE

km. 0 0.5 1 2 3 km.

January 1977

Resistivity in Ohm-meters

GT/149/74 (52931)

Figure 26

THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS – ISRAEL

ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 71

EMEQ YIZREEL

SCALE

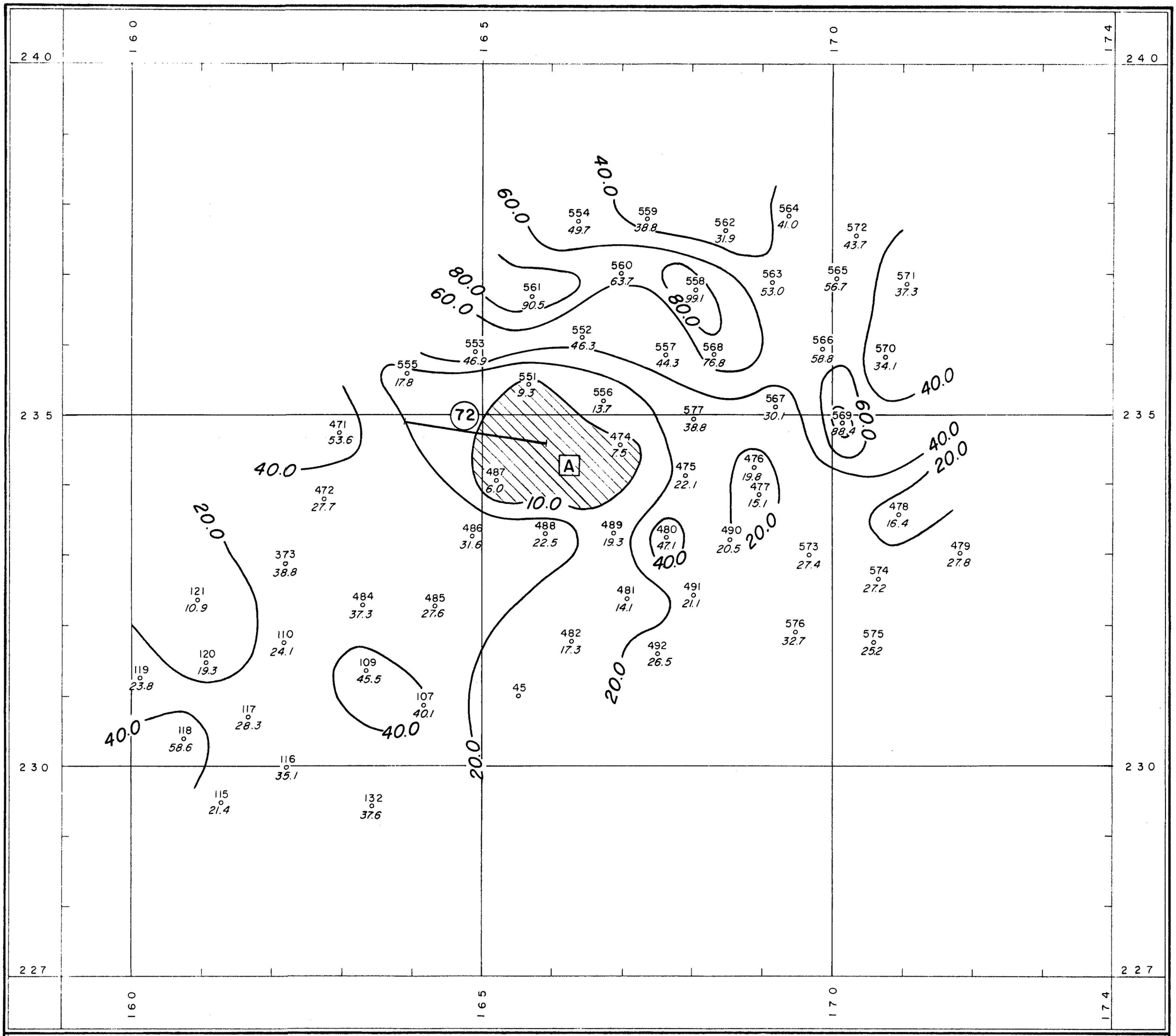
km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

January 1977

GE/149/74 (52953)

Figure 27



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 72

EMEQ YIZREEL

SCALE

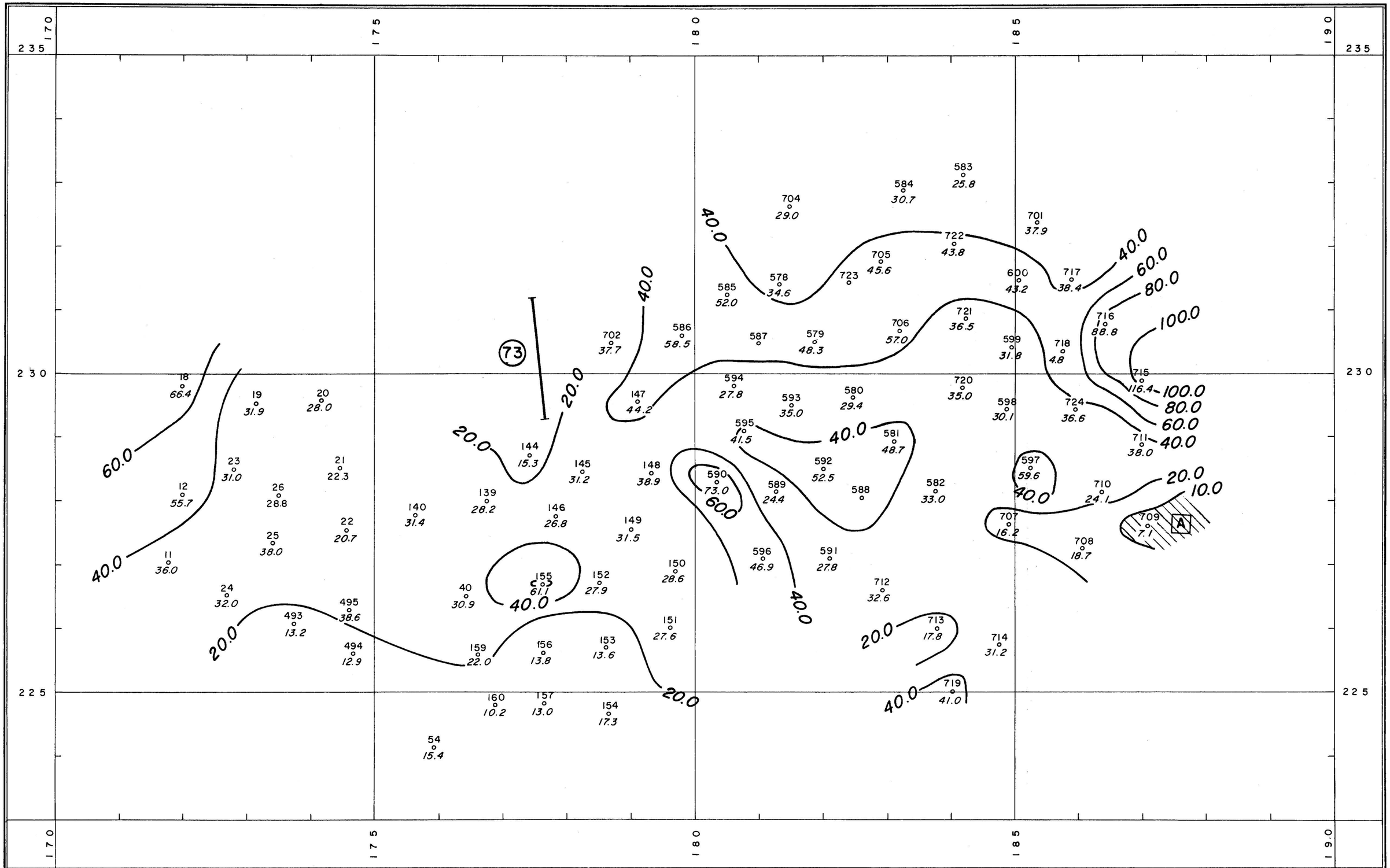
km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

January 1977

GE/149/74 (52954)

Figure 28



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 73

EMEQ YIZREEL

SCALE

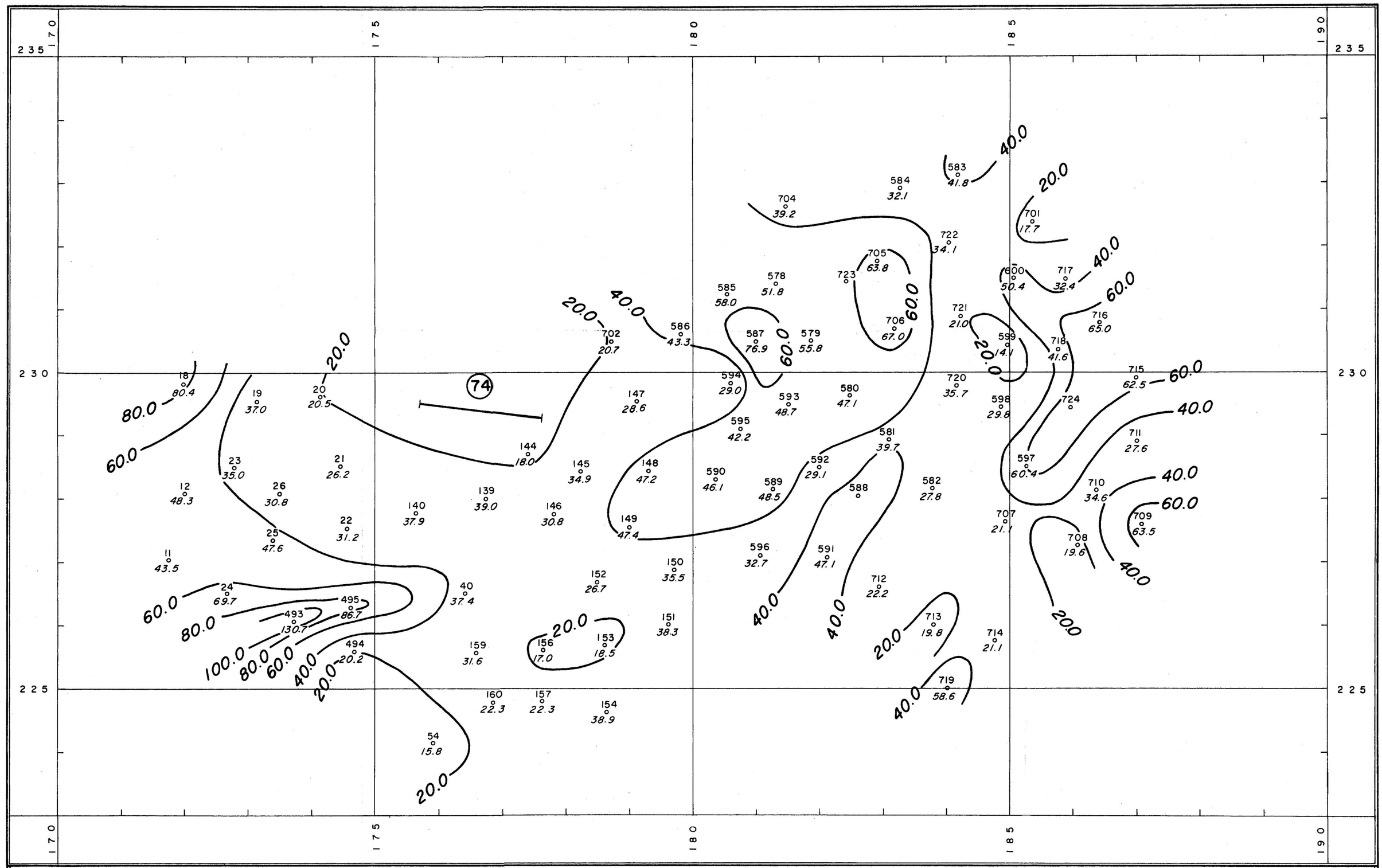
km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

January 1977

GE/149/74 (53955)

Figure 29



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 74

EMEQ YIZREEL

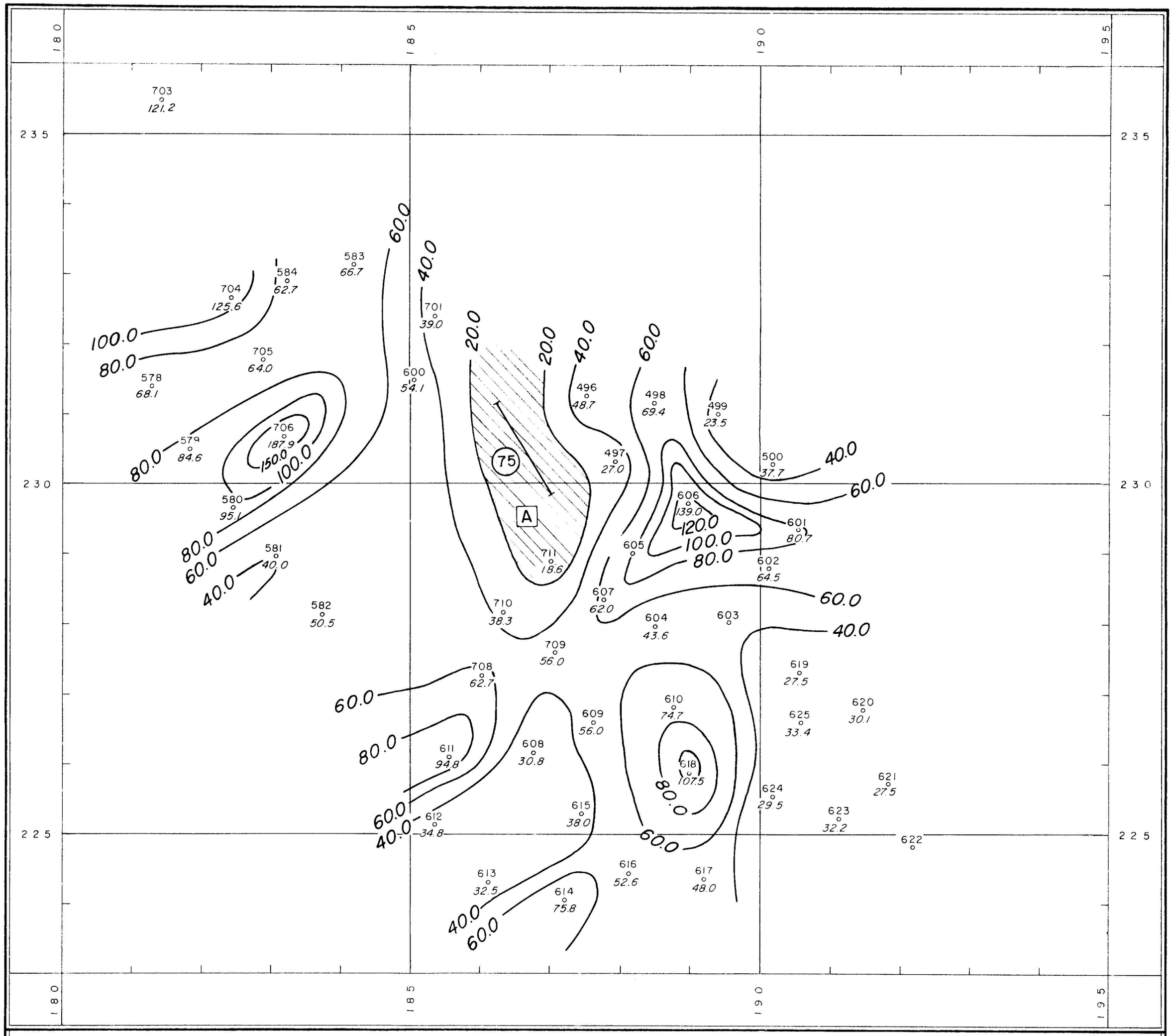
SCALE

km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

GE/149/74 (53956)

Figure 30



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 75

EMEQ YIZREEL

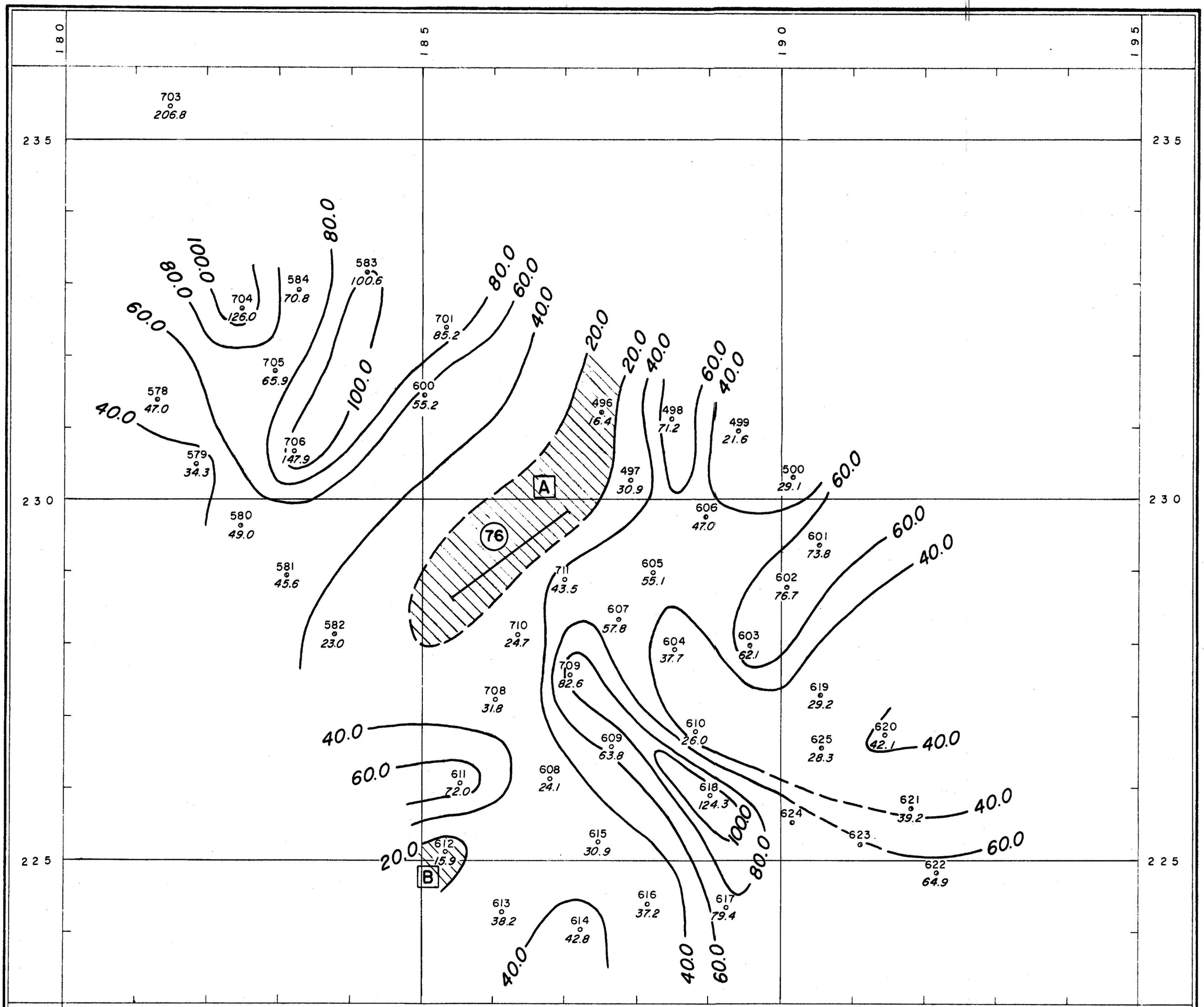
SCALE

km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

GE/149/74 (52957)

Figure 31



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 76

EMEQ YIZREEL

SCALE

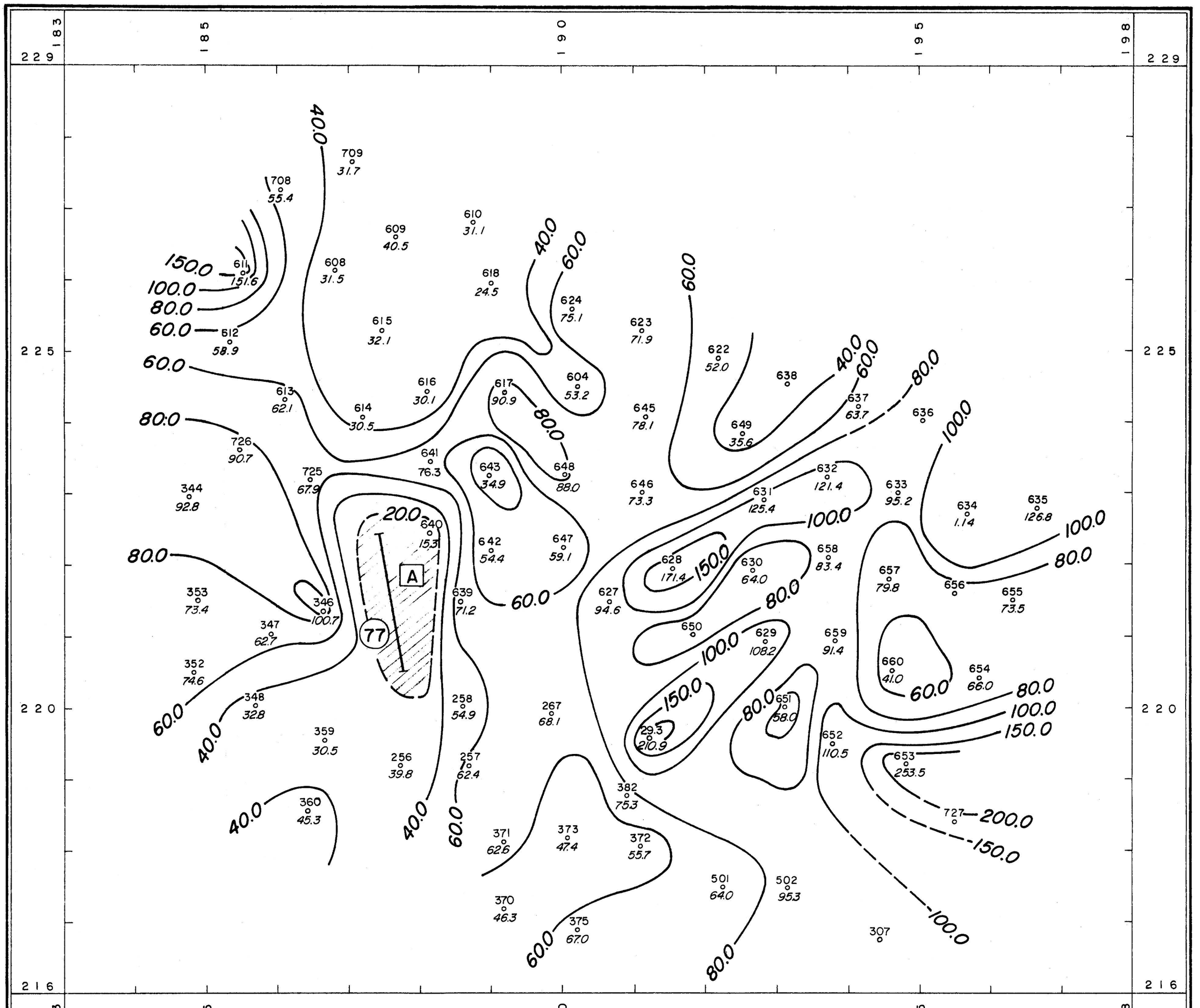
km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

January 1977

GE/149/74 (52958)

Figure 32



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 77

EMEQ YIZREEL

SCALE

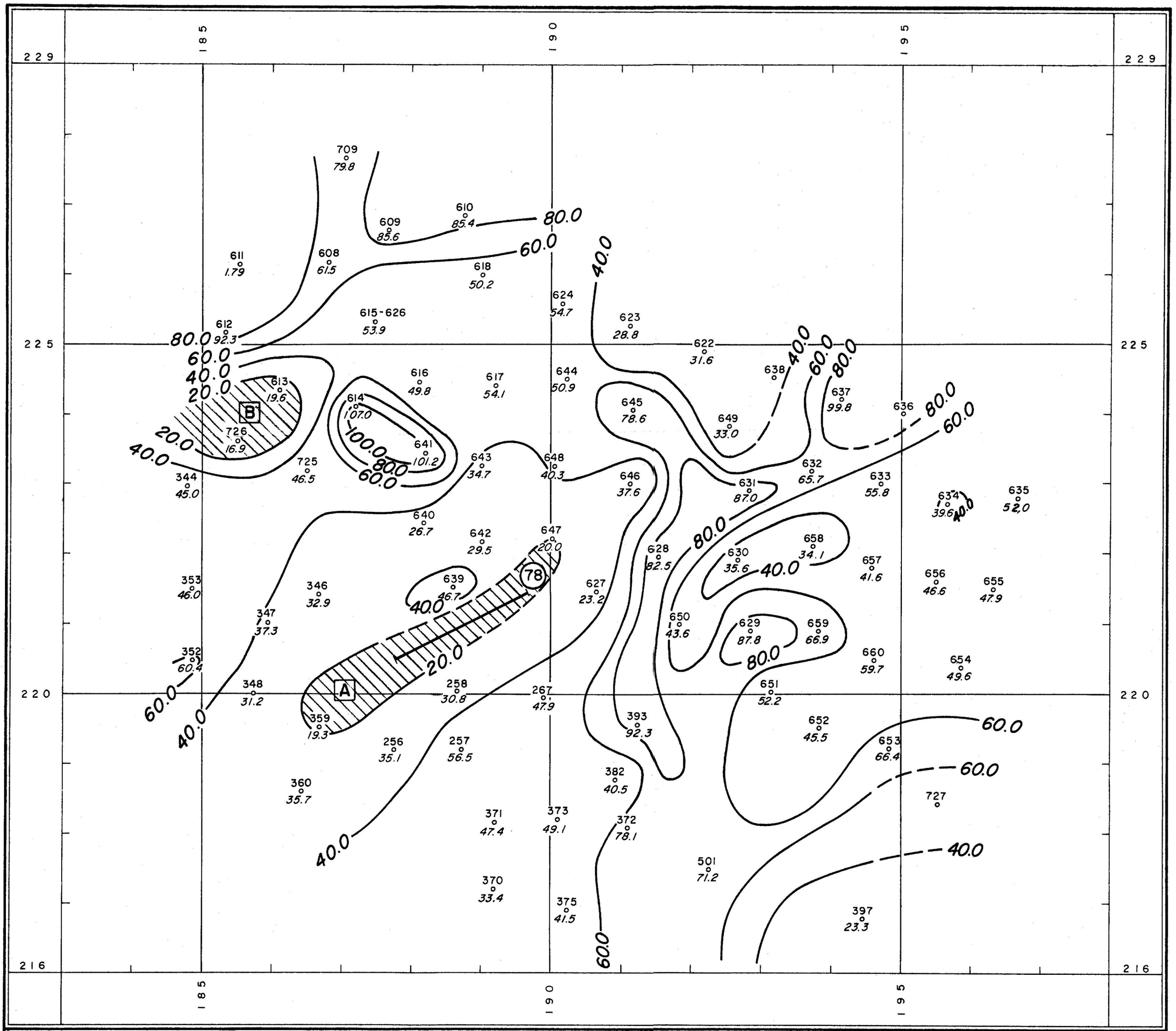
km. 0 0.5 1 2 3 km.

January 1977

Reststivity in Ohm-meters

GE/149/74 (53959)

Figure 33



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 78

EMEQ YIZREEL

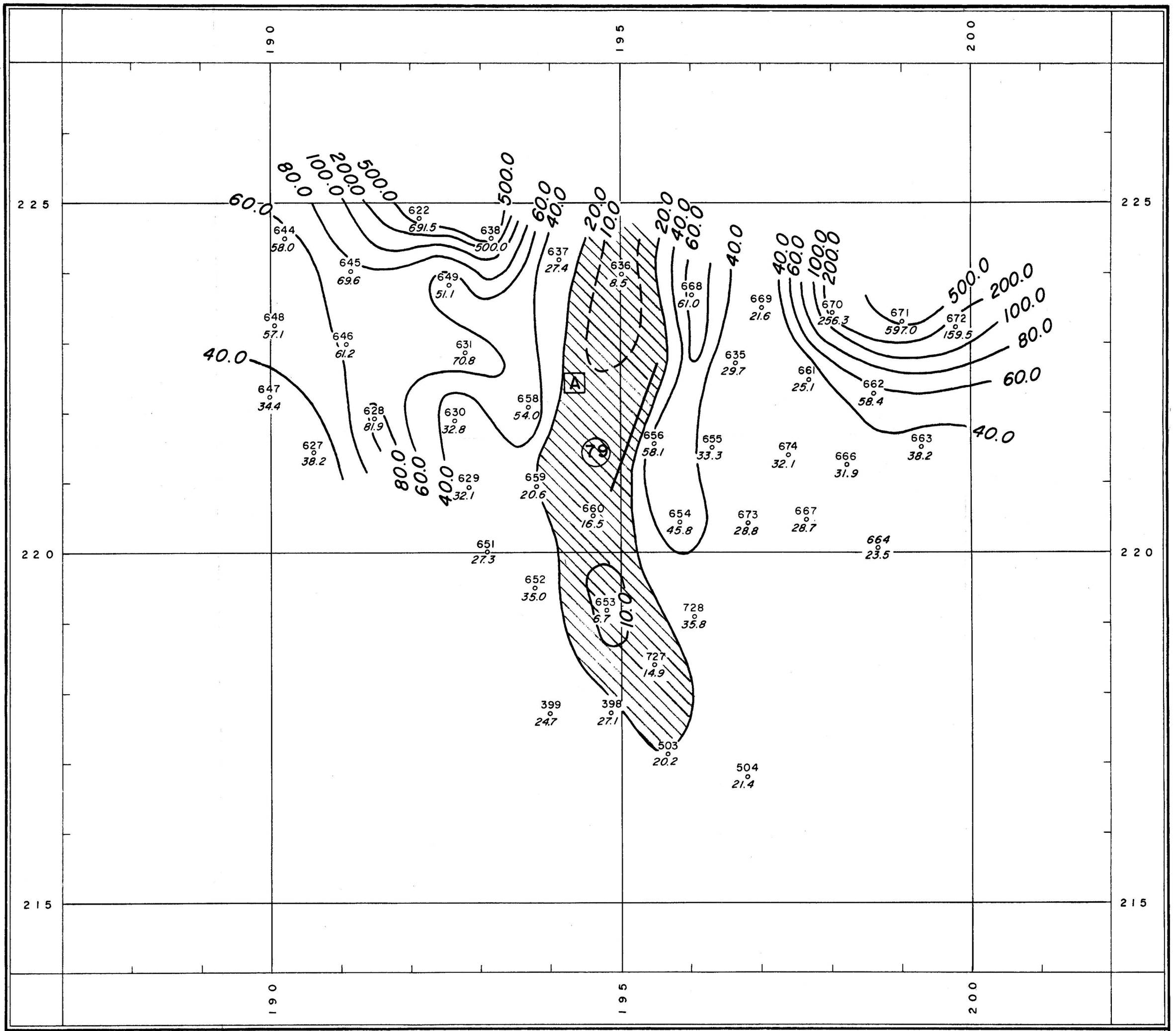
SCALE

km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

GE/149/74 (53960)

Figure 34



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 79

EMEQ YIZREEL

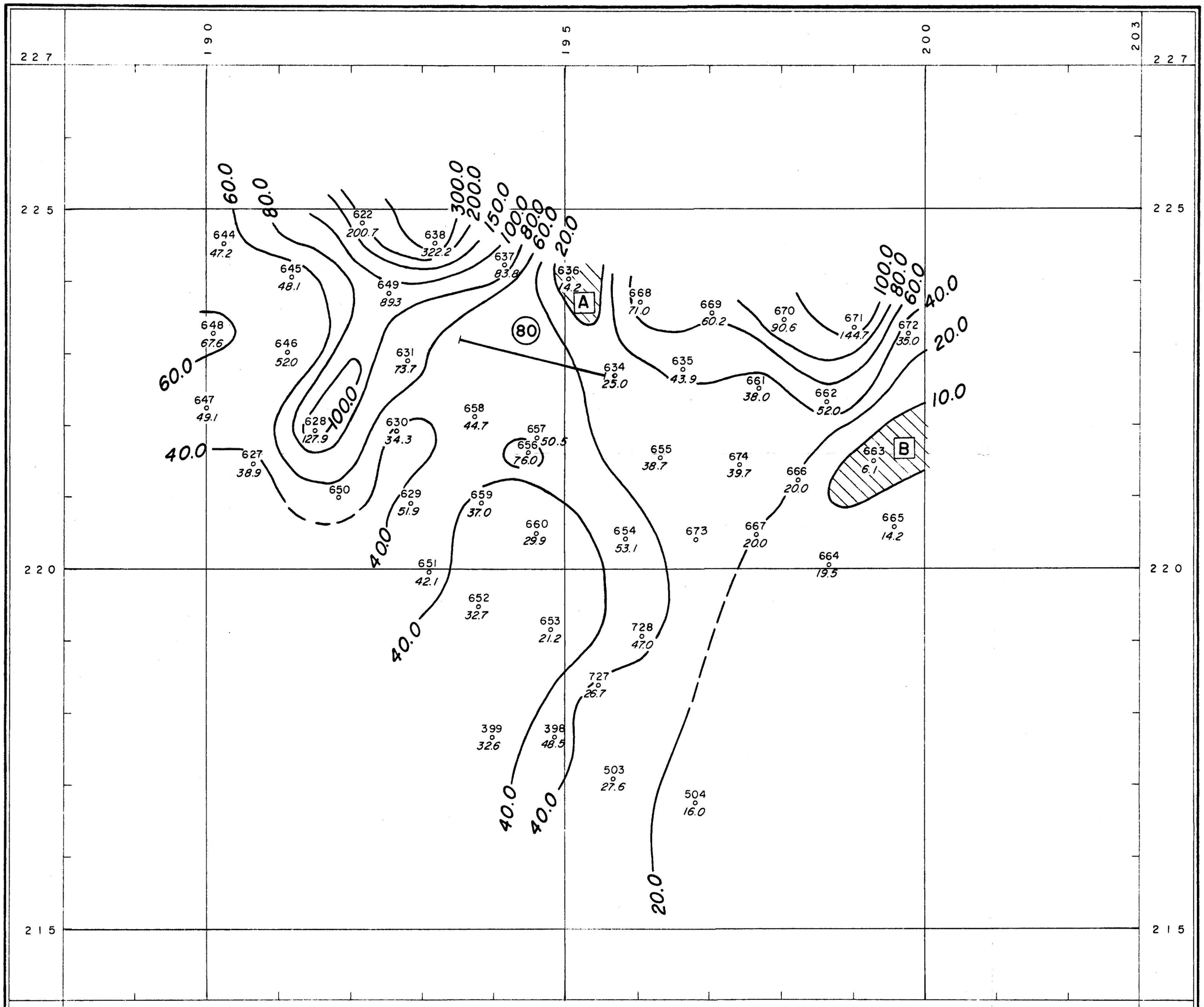
SCALE

km. 0 0.5 1 2 3 km.

Resistivity in Ohm - meters

GE/149/74 (53961)

Figure 35



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ROVING DIPOLE

APPARENT RESISTIVITY MAP

SOURCE 80

EMEQ YIZREEL

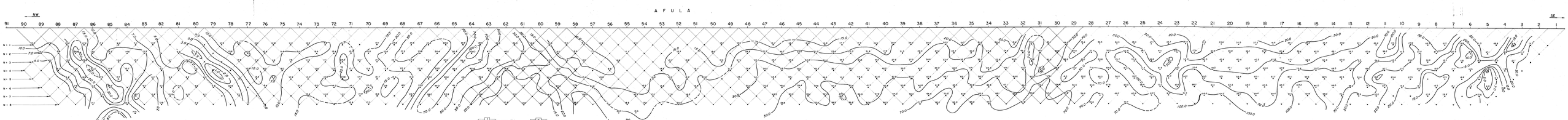
SCALE

km. 0 0.5 1 2 3 km.

Resistivity in Ohm-meters

G /149/74 (53962)

Figure 36


A F U L A
NW
SE
91 90 89 88 87 86 85 84 83 82 81 80 79 78 77 76 75 74 73 72 71 70 69 68 67 66 65 64 63 62 61 60 59 58 57 56 55 54 53 52 51 50 49 48 47 46 45 44 43 42 41 40 39 38 37 36 35 34 33 32 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1
N-1
N-2
N-3
N-4
N-5
N-6
N-7
N-8
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ISRAEL
DIPOLE - DIPOLE
APPARENT RESISTIVITY CROSS SECTION I
EMEQ YIZREEL
SCALE
PLOTTING POINT

Figure 37

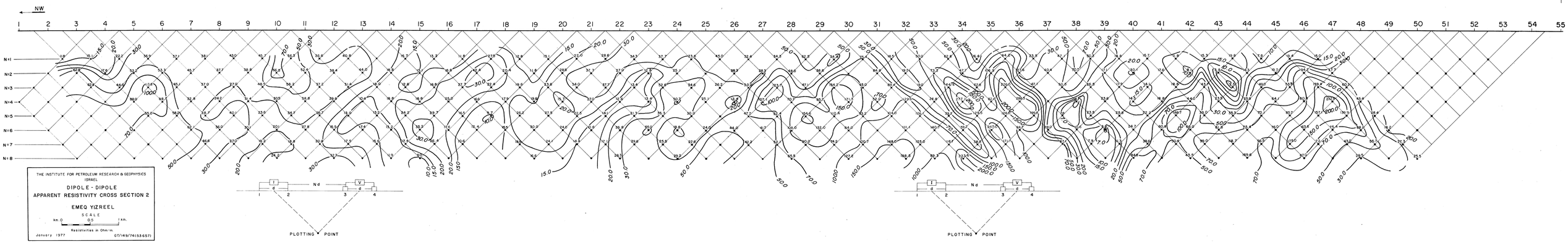
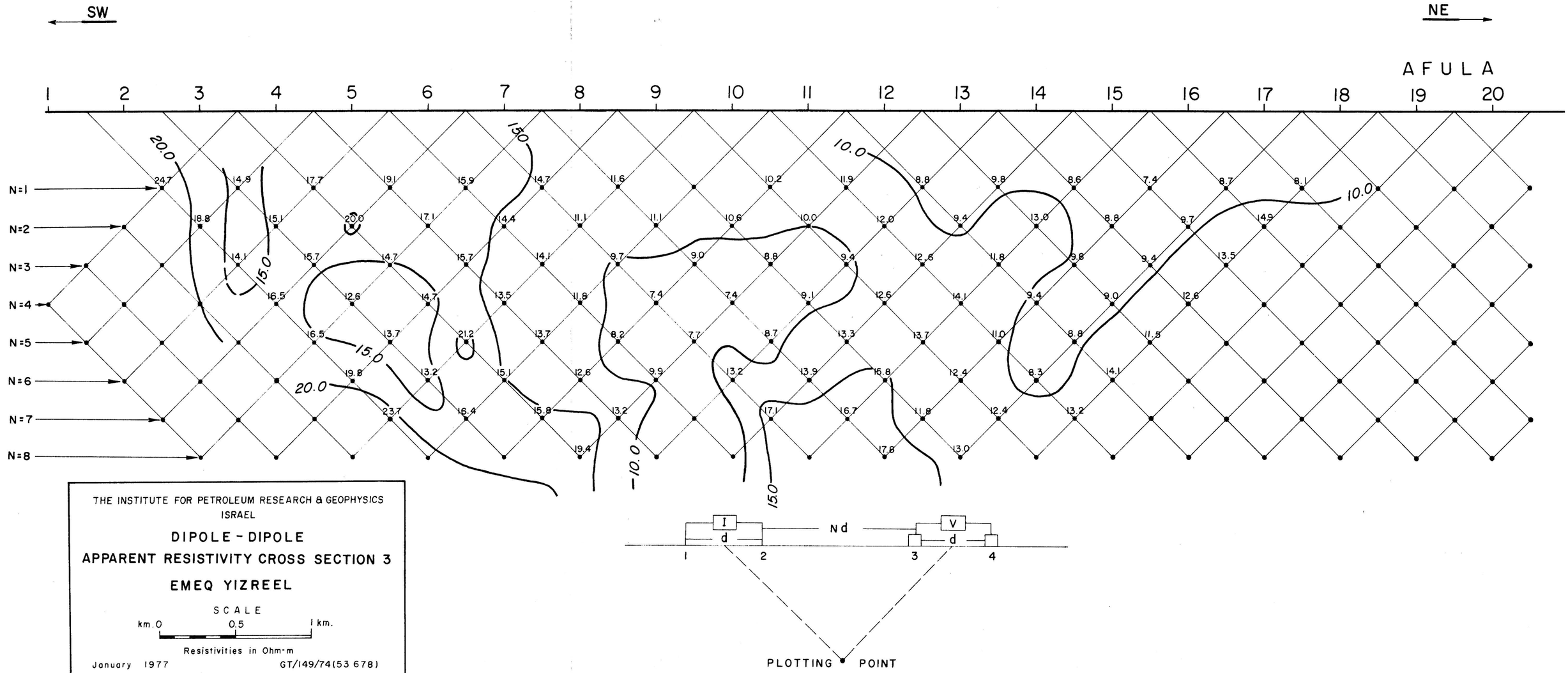

NW
SE
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55
N=1
N=2
N=3
N=4
N=5
N=6
N=7
N=8
THE INSTITUTE FOR PETROLEUM RESEARCH & GEOPHYSICS
ISRAEL
DIPOLE - DIPOLE
APPARENT RESISTIVITY CROSS SECTION 2
EMEQ YIZREEL
SCALE
km. 0 0.5 1 km.
Resistivities in Ohm-m
January 1977
07/149/74(53657)
I d 1 2 Nd V d 3 4
PLOTTING POINT
I d 1 2 Nd V d 3 4
PLOTTING POINT

Figure 38


SW
NE
AFULA
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
N=1
N=2
N=3
N=4
N=5
N=6
N=7
N=8
20.0
15.0
10.0
-10.0
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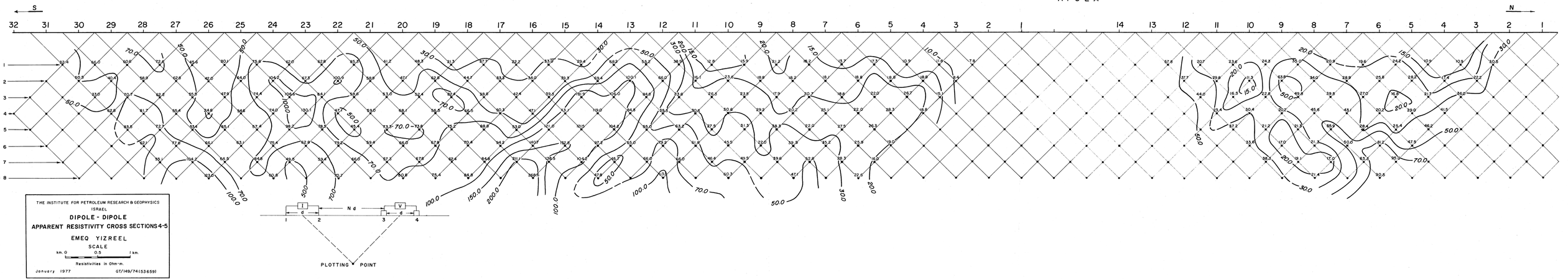
ISRAEL
DIPOLE - DIPOLE
APPARENT RESISTIVITY CROSS SECTION 3
EMEQ YIZREEL
SCALE
km. 0 0.5 1 km.
Resistivities in Ohm-m
January 1977
GT/149/74(53 678)
I
V
d
Nd
1 2 3 4
PLOTTING POINT

Figure 39



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DIPOLE - DIPOLE
APPARENT RESISTIVITY CROSS SECTIONS 4-5

EMEQ YIZREEL

SCALE
km. 0 0.5 1 km.

Resistivities in Ohm-m.

January 1977 GT/149/74(53659)

Figure 40

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COMPOSITE DIPOLE - DIPOLE MAP

APPARENT RESISTIVITIES FOR N = 4

EMEQ YIZREEL

Scale 1 : 50 000

km 0 1 2 3 4km

LEGEND

LOW APPARENT RESISTIVITY ZONE

APPARENT RESISTIVITY CONTOUR (Ohm-meters)

March 1977

GT/149/74 (53965)

Figure 41

NAZARETH

YOQNE'AM

AFULA

GID'ONA

JENIN

SECTION 1

SECTION 2

SECTION 3

SECTION 4

SECTION 5

SECTION 1

Figure 42

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INTERPRETATION MAP
ROVING DIPOLE LOW RESISTIVITY ZONES
EMEQ YIZREEL

Scale 1 : 50 000

km 0 1 2 3 4 km

LEGEND

- LOW APPARENT RESISTIVITY ZONE SOURCE 78 ANOMALY A — 78-A
- SOURCE LOCATION — 78
- WELL
- SPRING

March 1977

GT/149/74 (53963)